

Mathematical Prerequisites

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This is the single source for the analysis, linear algebra and measure-theoretic probability the course uses. Every result stated in a unit without proof is proved here.

What is assumed. The completeness of $\mathbb{R}$, and the Lebesgue integral on a measure space together with its four convergence theorems: monotone convergence, Fatou's lemma, dominated convergence, and Fubini–Tonelli. These are stated in §16 for reference and are the only unproved statements in the course. Everything above them is derived.

Part I

Analysis

1 The Real Line

Nothing beyond this section is assumed of analysis.

Definition P.1 (Supremum). s is the **supremum** of $S \subseteq \mathbb{R}$ if s is an upper bound and no smaller number is; $\inf S = -\sup(-S)$. The **completeness axiom** states that every non-empty S bounded above has a supremum in $\mathbb{R}$.

Theorem P.2 (Monotone convergence for sequences). A bounded increasing real sequence converges, to the supremum of its values.

Proof. Let $s = \sup_n a_n$, which exists by completeness. For $\varepsilon > 0$, $s - \varepsilon$ is not an upper bound, so $a_N > s - \varepsilon$ for some N ; monotonicity gives $s - \varepsilon < a_n \leq s$ for all $n \geq N$. $\square$

Theorem P.3 (Bolzano–Weierstrass). Every bounded sequence in $\mathbb{R}^d$ has a convergent subsequence.

Proof. Take $d = 1$ first. Given a bounded (a_n) , put $b_m = \sup_{n \geq m} a_n$, finite by completeness. The sequence (b_m) is decreasing and bounded below, so it converges to $b = \limsup_n a_n$ by Theorem P.2 applied to $(-b_m)$. Choose indices $n_1 < n_2 < \dots$ with $|a_{n_k} - b| < 1/k$: possible because $b_{n_{k-1}+1}$ is within $1/(2k)$ of b for large k , and being a supremum it is approached by some a_n with $n > n_{k-1}$. Then $a_{n_k} \rightarrow b$.

For $d > 1$ extract a subsequence convergent in the first coordinate, a further subsequence convergent in the second, and so on through the d coordinates; the last subsequence converges in every coordinate, hence in $\mathbb{R}^d$. $\square$

Definition P.4 (Compactness). $K \subseteq \mathbb{R}^d$ is **compact** if every sequence in K has a subsequence converging to a point of K .

Theorem P.5 (Heine–Borel). $K \subseteq \mathbb{R}^d$ is compact if and only if it is closed and bounded.

Proof. ($\Leftarrow$) A sequence in a bounded K has a convergent subsequence by Theorem P.3, and its limit lies in K because K is closed.

($\Rightarrow$) If K were unbounded it would contain x_n with $|x_n| > n$, and no subsequence of such a sequence converges. If K were not closed there would be $x_n \in K$ with $x_n \rightarrow x \notin K$; every subsequence also tends to x , so none converges in K . $\square$

Theorem P.6 (Rolle). If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then $f'(\xi) = 0$ for some $\xi \in (a, b)$.

Proof. $[a, b]$ is compact by Theorem P.5, so f attains a maximum and a minimum on it (Theorem P.14). If both are attained at endpoints then f is constant, being squeezed between two equal values, and any interior ξ works. Otherwise an extremum is attained at an interior ξ , where the difference quotients $\{f(\xi + h) - f(\xi)\}/h$ are ≤ 0 for $h > 0$ and ≥ 0 for $h < 0$ (or the reverse), so the two-sided limit $f'(\xi)$ is both ≤ 0 and ≥ 0 . $\square$

Corollary P.7 (Mean value theorem). Under the same hypotheses without $f(a) = f(b)$, there is $\xi \in (a, b)$ with $f(b) - f(a) = f'(\xi)(b - a)$.

Proof. Apply Theorem P.6 to $g(x) = f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x - a)$, which is continuous on $[a, b]$, differentiable on (a, b) , and has $g(a) = g(b) = 0$. $\square$

Theorem P.8 (Differentiation under the integral sign). Let $(\mathcal{X}, \mu)$ be a measure space, $I \subseteq \mathbb{R}$ an open interval, and $f : I \times \mathcal{X} \rightarrow \mathbb{R}$ such that

- (i) $x \mapsto f(t, x)$ is μ -integrable for each $t \in I$;
- (ii) $\partial f / \partial t$ exists for every (t, x) ;
- (iii) $|\partial f / \partial t(t, x)| \leq g(x)$ for all $t \in I$, with $\int g \, d\mu < \infty$.

Then $F(t) = \int f(t, x) \, d\mu(x)$ is differentiable on I with $F'(t) = \int \frac{\partial f}{\partial t}(t, x) \, d\mu(x)$.

Proof. Fix $t \in I$ and any sequence $h_n \rightarrow 0$ with $t + h_n \in I$, $h_n \neq 0$. Put $q_n(x) = \{f(t + h_n, x) - f(t, x)\} / h_n$. By (ii) and the mean value theorem (Corollary P.7) applied in t , for each x there is ξ between t and $t + h_n$ with $q_n(x) = \partial f / \partial t(\xi, x)$, so $|q_n(x)| \leq g(x)$ by (iii). Also

$q_n(x) \rightarrow \partial f / \partial t(t, x)$ pointwise by (ii). Dominated convergence gives $\int q_n d\mu \rightarrow \int \partial f / \partial t d\mu$, and the left side is $\{F(t + h_n) - F(t)\} / h_n$. As the sequence (h_n) was arbitrary, the limit defines $F'(t)$. $\square$

Remark. Theorem P.8 is what licenses the differentiation under the integral in the Gaussian characteristic function (Proposition P.82), where $f(t, z) = e^{itz}\phi(z)$ has $|\partial f / \partial t| = |z|\phi(z)$, integrable and free of t .

2 Limits and Semi-continuity

Definition P.9 (Upper and lower limits). For a real sequence (a_n) , $\limsup_n a_n = \inf_m \sup_{n \geq m} a_n$ and $\liminf_n a_n = \sup_m \inf_{n \geq m} a_n$, as elements of $[-\infty, \infty]$.

Proposition P.10. $\liminf_n a_n \leq \limsup_n a_n$, and $a_n \rightarrow a$ if and only if $\liminf_n a_n = \limsup_n a_n = a$.

Proof. $\inf_{n \geq m} a_n \leq \sup_{n \geq m} a_n$ for each m ; the left side increases in m and the right decreases, so the supremum of the former is at most the infimum of the latter. If both equal a then for $\varepsilon > 0$

there is m with $a - \varepsilon \leq \inf_{n \geq m} a_n$ and $\sup_{n \geq m} a_n \leq a + \varepsilon$, so $|a_n - a| \leq \varepsilon$ for $n \geq m$. The converse is immediate from the definitions. $\square$

Definition P.11 (Semi-continuity). $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is **lower semi-continuous** at x if $\liminf_{y \rightarrow x} f(y) \geq f(x)$, and **upper semi-continuous** if $\limsup_{y \rightarrow x} f(y) \leq f(x)$.

Lemma P.12 (Semi-continuous envelopes). For bounded $f : \mathbb{R}^d \rightarrow \mathbb{R}$ put $f_*(x) = \liminf_{y \rightarrow x} f(y)$ and $f^*(x) = \limsup_{y \rightarrow x} f(y)$. Then f_* is lower semi-continuous, f^* is upper semi-continuous, $f_* \leq f \leq f^*$, and the three agree exactly at the continuity points of f .

Proof. Fix x and $\delta > 0$. By definition of $\liminf$ there is a neighbourhood V of x with $f(y) \geq f_*(x) - \delta$ for all $y \in V$. For $z \in V$ every neighbourhood of z meets V , so $f_*(z) \geq f_*(x) - \delta$: that is lower semi-continuity. Applying this to $-f$ gives the claim for f^* . The inequalities and the agreement at continuity points are immediate. $\square$

Lemma P.13 (Approximation from below). A lower semi-continuous f bounded below is the pointwise increasing limit of the bounded continuous functions $f_k(x) = \inf_y \{f(y) \wedge k + k|x - y|\}$.

Proof. Each f_k is an infimum of functions that are k -Lipschitz in x , hence k -Lipschitz, hence continuous; it is bounded below by $\inf f$ and above by k . Taking $y = x$ gives $f_k \leq f \wedge k$, and $f_k \leq f_{k+1}$ since the bracket increases in k .

For the limit, fix x and $c < f(x)$. Lower semi-continuity gives $\delta > 0$ with $f(y) > c$ whenever $|y - x| < \delta$. For k with $k\delta > c - \inf f$ and $k > c$, split the infimum: if $|y - x| < \delta$ then $f(y) \wedge k + k|x - y| > c$, and if $|y - x| \geq \delta$ then $f(y) \wedge k + k|x - y| \geq \inf f + k\delta > c$. Hence $f_k(x) \geq c$, and $c < f(x)$ was arbitrary. $\square$

3 Continuity, Compactness, Differentiation

Theorem P.14 (Uniform continuity on compacts). A continuous f on a compact $K \subseteq \mathbb{R}^d$ (Definition P.4) is bounded, uniformly continuous, and attains its supremum.

Proof. Attainment. Let $M = \sup_K f$, finite or not, and take $x_n \in K$ with $f(x_n) \rightarrow M$. By compactness a subsequence converges to some $x \in K$, and continuity gives $f(x) = M$; in particular $M < \infty$, so f is bounded above, and applying this to $-f$ bounds it below.

Uniform continuity. If it failed there would be $\varepsilon > 0$ and points $x_n, y_n \in K$ with $|x_n - y_n| <$

$1/n$ but $|f(x_n) - f(y_n)| \geq \varepsilon$. Passing to a convergent subsequence $x_{n_k} \rightarrow x$ forces $y_{n_k} \rightarrow x$ too, and continuity makes both $f(x_{n_k})$ and $f(y_{n_k})$ tend to $f(x)$, contradicting the separation. $\square$

Theorem P.15 (Taylor with Lagrange remainder). If $f : \mathbb{R} \rightarrow \mathbb{R}$ is $m + 1$ times differentiable on an interval containing 0 and t , then for some ξ between 0 and t ,

$$f(t) = \sum_{k=0}^m \frac{f^{(k)}(0)}{k!} t^k + \frac{f^{(m+1)}(\xi)}{(m+1)!} t^{m+1}. \quad (\text{P.1})$$

Proof. Fix $t \neq 0$ and choose R so that $g(s) = f(t) - \sum_{k \leq m} \frac{f^{(k)}(s)}{k!} (t-s)^k - R(t-s)^{m+1}$ vanishes at $s = 0$; it also vanishes at $s = t$. By Rolle's theorem $g'(\xi) = 0$ for some ξ between. The sum telescopes on differentiation,

$$\frac{d}{ds} \sum_{k \leq m} \frac{f^{(k)}(s)}{k!} (t-s)^k = \frac{f^{(m+1)}(s)}{m!} (t-s)^m,$$

so $g'(\xi) = 0$ reads $-\frac{f^{(m+1)}(\xi)}{m!} (t-\xi)^m + R(m+1)(t-\xi)^m = 0$, giving $R = f^{(m+1)}(\xi)/(m+1)!$.

Substituting $s = 0$ into $g(0) = 0$ is (P.1). □

Lemma P.16 (Taylor bound for the complex exponential). For every $t \in \mathbb{R}$ and integer $m \geq 0$,

$$\left| e^{it} - \sum_{k=0}^m \frac{(it)^k}{k!} \right| \leq \min \left\{ \frac{2|t|^m}{m!}, \frac{|t|^{m+1}}{(m+1)!} \right\}. \quad (\text{P.2})$$

Proof. Let $h_m(t)$ be the difference without the modulus. Differentiating the series termwise, $h_m(t) = i \int_0^t h_{m-1}(s) ds$ for $m \geq 0$, with $|h_{-1}| \equiv 1$ and $|h_0(t)| = |e^{it} - 1| \leq 2$. If $|h_{m-1}(t)| \leq |t|^m/m!$ then $|h_m(t)| \leq \int_0^{|t|} s^m/m! ds = |t|^{m+1}/(m+1)!$; if $|h_{m-1}(t)| \leq 2|t|^{m-1}/(m-1)!$ then $|h_m(t)| \leq 2|t|^m/m!$. Both inductions start from the two displayed facts. □

Lemma P.17 (Product comparison). For complex $z_1, \dots, z_m$ and $z'_1, \dots, z'_m$ of modulus at most 1, $\left| \prod_k z_k - \prod_k z'_k \right| \leq \sum_k |z_k - z'_k|$.

Proof. For $m = 2$, inserting $z'_1 z_2$, $|z_1 z_2 - z'_1 z'_2| \leq |z_2| |z_1 - z'_1| + |z'_1| |z_2 - z'_2| \leq |z_1 - z'_1| + |z_2 - z'_2|$. Induct with $z_1 \rightsquigarrow \prod_{k < m} z_k$ and $z_2 \rightsquigarrow z_m$, both of modulus at most 1. □

4 Convexity and Separation

Unit 14 derives the support vector machine from Lagrangian duality; the geometry it needs is here.

Definition P.18 (Convex set, convex function). $C \subseteq \mathbb{R}^d$ is **convex** if $\lambda x + (1-\lambda)y \in C$ whenever $x, y \in C$ and $\lambda \in [0, 1]$. A function $f : C \rightarrow \mathbb{R}$ on a convex C is **convex** if $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$.

Theorem P.19 (Nearest point in a closed convex set). Let $C \subseteq \mathbb{R}^d$ be non-empty, closed and convex, and $z \notin C$. Then there is a unique $p \in C$ minimising $|z - x|$ over $x \in C$, and it is characterised by

$$\langle z - p, x - p \rangle \leq 0 \quad \text{for every } x \in C. \quad (\text{P.3})$$

Proof. Existence. Let $\delta = \inf_{x \in C} |z - x|$ and take $x_n \in C$ with $|z - x_n| \rightarrow \delta$. The parallelogram law gives

$$|x_n - x_m|^2 = 2|z - x_n|^2 + 2|z - x_m|^2 - 4\left|z - \frac{x_n + x_m}{2}\right|^2 \leq 2|z - x_n|^2 + 2|z - x_m|^2 - 4\delta^2,$$

using $\frac{1}{2}(x_n + x_m) \in C$ by convexity. The right side tends to 0, so (x_n) is Cauchy and converges to

some p , which lies in C because C is closed, with $|z - p| = \delta$.

Characterisation and uniqueness. For $x \in C$ and $\lambda \in (0, 1]$, the point $p + \lambda(x - p)$ lies in C , so

$$\delta^2 \leq |z - p - \lambda(x - p)|^2 = \delta^2 - 2\lambda\langle z - p, x - p \rangle + \lambda^2|x - p|^2.$$

Dividing by λ and letting $\lambda \rightarrow 0^+$ gives (P.3). Conversely if p satisfies (P.3) then for any $x \in C$, $|z - x|^2 = |z - p|^2 - 2\langle z - p, x - p \rangle + |x - p|^2 \geq |z - p|^2$, so p is minimising; and if p' also minimises, taking $x = p'$ in (P.3) for p and symmetrically gives $|p - p'|^2 \leq 0$. $\square$

Theorem P.20 (Separating hyperplane). Let $C \subseteq \mathbb{R}^d$ be non-empty, closed and convex, and $z \notin C$. Then there exist $\mathbf{w} \neq \mathbf{0}$ and $\alpha \in \mathbb{R}$ with $\langle \mathbf{w}, z \rangle > \alpha \geq \langle \mathbf{w}, x \rangle$ for every $x \in C$.

Proof. Let p be the nearest point of Theorem P.19 and put $\mathbf{w} = z - p \neq \mathbf{0}$. By (P.3), $\langle \mathbf{w}, x \rangle \leq \langle \mathbf{w}, p \rangle$ for every $x \in C$; take $\alpha = \langle \mathbf{w}, p \rangle$. Then $\langle \mathbf{w}, z \rangle - \alpha = \langle \mathbf{w}, z - p \rangle = |\mathbf{w}|^2 > 0$. $\square$

Proposition P.21 (First-order condition). A differentiable f on an open convex C is convex if and only if $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$ for all $x, y \in C$. A convex differentiable f attains a

global minimum at x if and only if $\nabla f(x) = \mathbf{0}$.

Proof. If f is convex then for $\lambda \in (0, 1]$, $f(x + \lambda(y - x)) \leq f(x) + \lambda\{f(y) - f(x)\}$, so

$$\frac{f(x + \lambda(y - x)) - f(x)}{\lambda} \leq f(y) - f(x);$$

letting $\lambda \rightarrow 0^+$ the left side tends to $\langle \nabla f(x), y - x \rangle$. Conversely, given the gradient inequality, put $u = \lambda x + (1 - \lambda)y$ and apply it at u to both x and y ; the combination $\lambda\{\cdot\} + (1 - \lambda)\{\cdot\}$ gives $\lambda f(x) + (1 - \lambda)f(y) \geq f(u)$, since the gradient terms cancel.

The minimum statement follows: $\nabla f(x) = \mathbf{0}$ makes the gradient inequality read $f(y) \geq f(x)$ for all y , and the converse is the usual first-order necessary condition on an open set. $\square$

Lemma P.22 (Coercive minimisation). Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be continuous and **coercive**, meaning $f(x) \rightarrow \infty$ as $|x| \rightarrow \infty$. Then f attains its infimum. If f is in addition **strictly convex**, that is $f(\lambda x + (1 - \lambda)y) < \lambda f(x) + (1 - \lambda)f(y)$ for $x \neq y$ and $\lambda \in (0, 1)$, the minimiser is unique.

Proof. Fix x_0 and put $S = \{x : f(x) \leq f(x_0)\}$, which contains x_0 and is closed by continuity. Coercivity gives R with $f(x) > f(x_0)$ whenever $|x| > R$, so S is bounded, hence compact by Theorem P.5. By Theorem P.14 f attains a minimum on S at some x^* , and for $x \notin S$ one has $f(x) > f(x_0) \geq f(x^*)$, so x^* minimises f on $\mathbb{R}^d$.

If $x^* \neq y^*$ both minimise, with common value m , strict convexity gives $f(\frac{1}{2}(x^* + y^*)) < m$, contradicting minimality. $\square$

Lemma P.23 (Directional characterisation of a convex minimum). Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be convex. For each x and v the difference quotient $t \mapsto \{f(x + tv) - f(x)\}/t$ is non-decreasing on $t > 0$, so

$$f'(x; v) = \lim_{t \downarrow 0} \frac{f(x + tv) - f(x)}{t} \quad (\text{P.4})$$

exists in $[-\infty, \infty)$. Moreover x minimises f over $\mathbb{R}^d$ if and only if $f'(x; v) \geq 0$ for every v .

Proof. For $0 < s < t$ write $x + sv = (1 - s/t)x + (s/t)(x + tv)$; convexity gives $f(x + sv) \leq (1 - s/t)f(x) + (s/t)f(x + tv)$, and subtracting $f(x)$ and dividing by s is the monotonicity. A

non-decreasing function has a limit from the right, finite or $-\infty$, and the limit is bounded above by the quotient at any $t > 0$, whence

$$f(x + tv) \geq f(x) + t f'(x; v) \quad (t > 0). \quad (\text{P.5})$$

If $f'(x; v) \geq 0$ for every v , then (P.5) with $t = 1$ and $v = y - x$ gives $f(y) \geq f(x)$ for every y . Conversely, if x is a minimiser, every difference quotient in (P.4) is non-negative, hence so is the limit. $\square$

Theorem P.24 (Separation of two convex sets). Let $C, D \subseteq \mathbb{R}^d$ be non-empty disjoint convex sets. Then there are $\mathbf{w} \neq \mathbf{0}$ and α with $\langle \mathbf{w}, x \rangle \leq \alpha \leq \langle \mathbf{w}, y \rangle$ for all $x \in C, y \in D$.

Proof. $E = C - D = \{x - y : x \in C, y \in D\}$ is convex, being a sum of convex sets, and $\mathbf{0} \notin E$ because C and D are disjoint.

If $\mathbf{0} \notin \overline{E}$, Theorem P.20 applied to the closed convex $\overline{E}$ and the point $\mathbf{0}$ gives $\mathbf{w}$ with $\langle \mathbf{w}, z \rangle \leq \langle \mathbf{w}, \mathbf{0} \rangle = 0$ for all $z \in \overline{E}$, after replacing $\mathbf{w}$ by $-\mathbf{w}$.

If $\mathbf{0} \in \overline{E} \setminus E$, choose $z_k \rightarrow \mathbf{0}$ with $z_k \notin \overline{E}$, possible since $\mathbf{0}$ is not interior to E : were every

neighbourhood of $\mathbf{0}$ inside $\overline{E}$, convexity would place $\mathbf{0}$ in E . For each k , Theorem P.20 gives a unit vector $\mathbf{w}_k$ with $\langle \mathbf{w}_k, z \rangle \leq \langle \mathbf{w}_k, z_k \rangle$ for all $z \in \overline{E}$. The $\mathbf{w}_k$ lie on the compact unit sphere, so by Theorem P.3 a subsequence converges to a unit $\mathbf{w}$, and passing to the limit gives $\langle \mathbf{w}, z \rangle \leq 0$ for all $z \in E$.

Either way $\langle \mathbf{w}, x - y \rangle \leq 0$ for $x \in C$, $y \in D$, so $\sup_{x \in C} \langle \mathbf{w}, x \rangle \leq \inf_{y \in D} \langle \mathbf{w}, y \rangle$; take α between them. $\square$

5 Convex Programming and Duality

Unit 14 derives the support vector machine as a convex program. Consider

$$p^* = \inf f_0(x) \quad \text{subject to} \quad f_i(x) \leq 0, \quad i = 1, \dots, m, \quad x \in \mathcal{D} \subseteq \mathbb{R}^d, \quad (\text{P.6})$$

with $\mathcal{D}$ convex and $f_0, \dots, f_m$ convex on $\mathcal{D}$.

Definition P.25 (Lagrangian and dual function). For $\boldsymbol{\lambda} \in \mathbb{R}^m$,

$$L(x, \boldsymbol{\lambda}) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x), \quad g(\boldsymbol{\lambda}) = \inf_{x \in \mathcal{D}} L(x, \boldsymbol{\lambda}), \quad (\text{P.7})$$

and the **dual problem** is $d^* = \sup_{\boldsymbol{\lambda} \geq \mathbf{0}} g(\boldsymbol{\lambda})$.

Proposition P.26 (Concavity of the dual). g is concave on $\mathbb{R}^m$, whatever the f_i .

Proof. For each fixed x , $\boldsymbol{\lambda} \mapsto L(x, \boldsymbol{\lambda})$ is affine. An infimum of affine functions is concave: if $g(\boldsymbol{\lambda}) = \inf_x L(x, \boldsymbol{\lambda})$ then for $\theta \in [0, 1]$,

$$L(x, \theta \boldsymbol{\lambda} + (1 - \theta) \boldsymbol{\mu}) = \theta L(x, \boldsymbol{\lambda}) + (1 - \theta) L(x, \boldsymbol{\mu}) \geq \theta g(\boldsymbol{\lambda}) + (1 - \theta) g(\boldsymbol{\mu}),$$

and taking the infimum over x on the left preserves the bound. □

Theorem P.27 (Weak duality). $d^* \leq p^*$.

Proof. Let x be feasible and $\boldsymbol{\lambda} \geq \mathbf{0}$. Then $\lambda_i f_i(x) \leq 0$ for every i , so $L(x, \boldsymbol{\lambda}) \leq f_0(x)$ and hence $g(\boldsymbol{\lambda}) \leq L(x, \boldsymbol{\lambda}) \leq f_0(x)$. Taking the infimum over feasible x gives $g(\boldsymbol{\lambda}) \leq p^*$, and then the supremum over $\boldsymbol{\lambda} \geq \mathbf{0}$. □

Theorem P.28 (Slater implies strong duality). Suppose p^* is finite and **Slater's condition** holds: there is $\tilde{x} \in \mathcal{D}$ with $f_i(\tilde{x}) < 0$ for every i . Then $d^* = p^*$, and the dual optimum is attained.

Proof. Put

$$\mathcal{A} = \{(\mathbf{u}, t) \in \mathbb{R}^m \times \mathbb{R} : \exists x \in \mathcal{D} \text{ with } f_i(x) \leq u_i \ \forall i, \ f_0(x) \leq t\}.$$

$\mathcal{A}$ is convex: given $(\mathbf{u}, t)$ from x and $(\mathbf{u}', t')$ from x' , convexity of $\mathcal{D}$ and of each f_i makes $\theta x + (1 - \theta)x'$ witness $\theta(\mathbf{u}, t) + (1 - \theta)(\mathbf{u}', t')$. Also $\mathcal{B} = \{(\mathbf{0}, s) : s < p^*\}$ is convex, and $\mathcal{A} \cap \mathcal{B} = \emptyset$, since a point of the intersection would be a feasible x with $f_0(x) \leq s < p^*$.

By Theorem P.24 there are $(\boldsymbol{\lambda}, \mu) \neq \mathbf{0}$ and α with

$$\langle \boldsymbol{\lambda}, \mathbf{u} \rangle + \mu t \geq \alpha \text{ on } \mathcal{A}, \quad \mu s \leq \alpha \text{ for all } s < p^*. \quad (\text{P.8})$$

Increasing any u_i or t keeps a point in $\mathcal{A}$, so $\boldsymbol{\lambda} \geq \mathbf{0}$ and $\mu \geq 0$; and letting $s \uparrow p^*$ gives $\mu p^* \leq \alpha$.

Case $\mu > 0$. Dividing (P.8) by μ and taking $(\mathbf{u}, t) = (f_1(x), \dots, f_m(x), f_0(x)) \in \mathcal{A}$ for arbitrary $x \in \mathcal{D}$,

$$f_0(x) + \sum_i \frac{\lambda_i}{\mu} f_i(x) \geq \frac{\alpha}{\mu} \geq p^*,$$

so $g(\boldsymbol{\lambda}/\mu) \geq p^*$. With Theorem P.27 this forces $d^* = p^*$, attained at $\boldsymbol{\lambda}/\mu$.

Case $\mu = 0$. Then $\langle \boldsymbol{\lambda}, \mathbf{u} \rangle \geq \alpha \geq \mu p^* = 0$ on $\mathcal{A}$, so $\sum_i \lambda_i f_i(x) \geq 0$ for every $x \in \mathcal{D}$. Taking $x = \tilde{x}$ makes every $f_i(\tilde{x}) < 0$, which with $\boldsymbol{\lambda} \geq \mathbf{0}$ forces $\boldsymbol{\lambda} = \mathbf{0}$, contradicting $(\boldsymbol{\lambda}, \mu) \neq \mathbf{0}$. So this case cannot occur. $\square$

Theorem P.29 (Karush–Kuhn–Tucker). Assume $f_0, \dots, f_m$ differentiable and strong duality, with x^* primal and $\boldsymbol{\lambda}^*$ dual optimal. Then

$$f_i(x^*) \leq 0, \quad \lambda_i^* \geq 0, \quad (\text{P.9})$$

$$\lambda_i^* f_i(x^*) = 0 \quad (i = 1, \dots, m), \quad \nabla f_0(x^*) + \sum_i \lambda_i^* \nabla f_i(x^*) = \mathbf{0}. \quad (\text{P.10})$$

Conversely, if the problem is convex and $(x^*, \boldsymbol{\lambda}^*)$ satisfies (P.9)–(P.10), then they are primal and dual optimal with zero gap.

Proof. Necessity. Strong duality gives

$$f_0(x^*) = g(\boldsymbol{\lambda}^*) = \inf_{x \in \mathcal{D}} \left\{ f_0(x) + \sum_i \lambda_i^* f_i(x) \right\} \leq f_0(x^*) + \sum_i \lambda_i^* f_i(x^*) \leq f_0(x^*),$$

the last step because each $\lambda_i^* f_i(x^*) \leq 0$. So both inequalities are equalities. Equality on the right forces $\sum_i \lambda_i^* f_i(x^*) = 0$, a sum of non-positive terms, hence each term is zero: that is **complementary slackness**. Equality on the left says x^* minimises $x \mapsto L(x, \boldsymbol{\lambda}^*)$ over $\mathcal{D}$, so its gradient vanishes there.

Sufficiency. Complementary slackness gives $L(x^*, \boldsymbol{\lambda}^*) = f_0(x^*)$. Under convexity, $L(\cdot, \boldsymbol{\lambda}^*)$ is convex with vanishing gradient at x^* , so by Proposition P.21 x^* minimises it, i.e. $g(\boldsymbol{\lambda}^*) = L(x^*, \boldsymbol{\lambda}^*) = f_0(x^*)$. Since x^* is feasible and $\boldsymbol{\lambda}^* \geq \mathbf{0}$, weak duality closes the gap. $\square$

Remark. Complementary slackness is what makes support vectors: $\lambda_i^* > 0$ forces $f_i(x^*) = 0$, so only the observations whose margin constraint is active carry a nonzero multiplier, and the solution depends on those alone.

Part II

Linear Algebra

6 Rank, Column Space, Rank–Nullity

For $\mathbf{A} \in \mathbb{R}^{n \times p}$ write $\mathcal{C}(\mathbf{A}) = \{\mathbf{A}\mathbf{v} : \mathbf{v} \in \mathbb{R}^p\}$ for the column space and $\mathcal{N}(\mathbf{A}) = \{\mathbf{v} : \mathbf{A}\mathbf{v} = \mathbf{0}\}$ for the null space, and $\text{rank}(\mathbf{A}) = \dim \mathcal{C}(\mathbf{A})$.

Theorem P.30 (Rank–nullity). $\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A}) = p$.

Proof. Take a basis $\mathbf{v}_1, \dots, \mathbf{v}_q$ of $\mathcal{N}(\mathbf{A})$ and extend it to a basis $\mathbf{v}_1, \dots, \mathbf{v}_p$ of $\mathbb{R}^p$. The vectors $\mathbf{A}\mathbf{v}_{q+1}, \dots, \mathbf{A}\mathbf{v}_p$ span $\mathcal{C}(\mathbf{A})$, since $\mathbf{A}\mathbf{v}_j = \mathbf{0}$ for $j \leq q$. They are independent: if $\sum_{j>q} c_j \mathbf{A}\mathbf{v}_j = \mathbf{0}$ then $\sum_{j>q} c_j \mathbf{v}_j \in \mathcal{N}(\mathbf{A})$, so it is a combination of $\mathbf{v}_1, \dots, \mathbf{v}_q$, and independence of the full basis forces every $c_j = 0$. Hence $\text{rank}(\mathbf{A}) = p - q$. $\square$

Proposition P.31 (Gram matrices). $\mathcal{N}(\mathbf{A}'\mathbf{A}) = \mathcal{N}(\mathbf{A})$ and $\text{rank}(\mathbf{A}'\mathbf{A}) = \text{rank}(\mathbf{A})$. In particular $\mathbf{A}'\mathbf{A}$ is invertible if and only if $\mathbf{A}$ has full column rank.

Proof. If $\mathbf{A}\mathbf{v} = \mathbf{0}$ then $\mathbf{A}'\mathbf{A}\mathbf{v} = \mathbf{0}$. Conversely $\mathbf{A}'\mathbf{A}\mathbf{v} = \mathbf{0}$ gives $\mathbf{v}'\mathbf{A}'\mathbf{A}\mathbf{v} = |\mathbf{A}\mathbf{v}|^2 = 0$, so $\mathbf{A}\mathbf{v} = \mathbf{0}$. The null spaces coincide, and both matrices have p columns, so Theorem P.30 equates the ranks. $\mathbf{A}'\mathbf{A}$ is $p \times p$, so it is invertible exactly when its rank is p . $\square$

7 The Trace

Definition P.32 (Trace). $\text{tr}(\mathbf{A}) = \sum_i a_{ii}$ for square $\mathbf{A}$.

Proposition P.33 (Properties).

- (i) tr is linear: $\text{tr}(\alpha\mathbf{A} + \beta\mathbf{B}) = \alpha \text{tr}(\mathbf{A}) + \beta \text{tr}(\mathbf{B})$.
- (ii) $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ whenever both products are defined, even when $\mathbf{AB}$ and $\mathbf{BA}$ have different sizes.
- (iii) $\text{tr}(\mathbf{PAP}^{-1}) = \text{tr}(\mathbf{A})$ for invertible $\mathbf{P}$: the trace is a similarity invariant.
- (iv) $\text{tr}(\mathbf{A}'\mathbf{A}) = \sum_{i,j} a_{ij}^2$, so $\text{tr}(\mathbf{A}'\mathbf{A}) = 0$ forces $\mathbf{A} = \mathbf{0}$.

Proof. (i) is immediate. For (ii), with $\mathbf{A}$ of size $m \times n$ and $\mathbf{B}$ of size $n \times m$,

$$\operatorname{tr}(\mathbf{AB}) = \sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ji} = \sum_{j=1}^n \sum_{i=1}^m b_{ji} a_{ij} = \operatorname{tr}(\mathbf{BA}),$$

the two double sums differing only in the order of summation, which is finite. (iii) follows by applying (ii) with the pair $\mathbf{PA}$ and $\mathbf{P}^{-1}$: $\operatorname{tr}((\mathbf{PA})\mathbf{P}^{-1}) = \operatorname{tr}(\mathbf{P}^{-1}(\mathbf{PA})) = \operatorname{tr}(\mathbf{A})$. For (iv), the (i, i) entry of $\mathbf{A}'\mathbf{A}$ is $\sum_j a_{ji}^2$. $\square$

Remark. Property (iii) is what makes $\operatorname{tr}(\mathbf{A})$ well defined on a linear map rather than on a matrix, and it is the step used in Lemma P.43 to pass from $\mathbf{PAP}^{-1}$ to $\mathbf{A}$. Property (ii) is used in Lemma P.60.

8 Inner Products, Orthogonality, Projection

Theorem P.34 (Cauchy–Schwarz). $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq |\mathbf{u}| |\mathbf{v}|$, with equality iff $\mathbf{u}$ and $\mathbf{v}$ are proportional.

Proof. For $\mathbf{v} \neq \mathbf{0}$ and $t \in \mathbb{R}$, $0 \leq |\mathbf{u} - t\mathbf{v}|^2 = |\mathbf{u}|^2 - 2t\langle \mathbf{u}, \mathbf{v} \rangle + t^2|\mathbf{v}|^2$. The right side is a quadratic in t with non-negative values, so its discriminant $4\langle \mathbf{u}, \mathbf{v} \rangle^2 - 4|\mathbf{u}|^2|\mathbf{v}|^2$ is at most 0. Equality forces a double root t_0 , at which $|\mathbf{u} - t_0\mathbf{v}| = 0$. $\square$

Theorem P.35 (Orthogonal projection). Let $S \subseteq \mathbb{R}^n$ be a subspace and $\mathbf{y} \in \mathbb{R}^n$. There is a unique $\hat{\mathbf{y}} \in S$ minimising $|\mathbf{y} - \mathbf{u}|$ over $\mathbf{u} \in S$, characterised by $\mathbf{y} - \hat{\mathbf{y}} \perp S$. If $S = \mathcal{C}(\mathbf{X})$ with $\mathbf{X}$ of full column rank, then $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ with $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

Proof. A subspace is closed and convex, so Theorem P.19 gives existence and uniqueness; and for a subspace the variational inequality (P.3) holds for $\mathbf{u}$ and for $2\hat{\mathbf{y}} - \mathbf{u}$, forcing equality, i.e. $\langle \mathbf{y} - \hat{\mathbf{y}}, \mathbf{u} - \hat{\mathbf{y}} \rangle = 0$ for every $\mathbf{u} \in S$, which is $\mathbf{y} - \hat{\mathbf{y}} \perp S$. For the formula, $\mathbf{y} - \hat{\mathbf{y}} \perp \mathcal{C}(\mathbf{X})$ means $\mathbf{X}'(\mathbf{y} - \hat{\mathbf{y}}) = \mathbf{0}$; writing $\hat{\mathbf{y}} = \mathbf{X}\mathbf{c}$ gives $\mathbf{X}'\mathbf{X}\mathbf{c} = \mathbf{X}'\mathbf{y}$, and $\mathbf{X}'\mathbf{X}$ is invertible by Proposition P.31. $\square$

Theorem P.36 (Pythagoras). If $\mathbf{u} \perp \mathbf{v}$ then $|\mathbf{u} + \mathbf{v}|^2 = |\mathbf{u}|^2 + |\mathbf{v}|^2$.

Proof. $|\mathbf{u} + \mathbf{v}|^2 = |\mathbf{u}|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + |\mathbf{v}|^2$. $\square$

9 The Spectral Theorem

Theorem P.37 (Spectral theorem). A symmetric $\mathbf{A} \in \mathbb{R}^{n \times n}$ has an orthonormal basis of eigenvectors: there are an orthogonal $\mathbf{P}$ and a diagonal $\mathbf{\Lambda}$ with $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$, the diagonal entries of $\mathbf{\Lambda}$ being the eigenvalues of $\mathbf{A}$, all real.

Proof. Induct on n . The Rayleigh quotient $\mathbf{v} \mapsto \mathbf{v}'\mathbf{A}\mathbf{v}$ is continuous on the unit sphere, which is compact, so by Theorem P.14 it attains its maximum λ_1 at some $\mathbf{v}_1$ with $|\mathbf{v}_1| = 1$. For any $\mathbf{u} \perp \mathbf{v}_1$ and $t \in \mathbb{R}$, put $\mathbf{w}(t) = (\mathbf{v}_1 + t\mathbf{u})/|\mathbf{v}_1 + t\mathbf{u}|$; then $t \mapsto \mathbf{w}(t)'\mathbf{A}\mathbf{w}(t)$ is differentiable with a maximum at $t = 0$, and

$$\left. \frac{d}{dt} \frac{(\mathbf{v}_1 + t\mathbf{u})'\mathbf{A}(\mathbf{v}_1 + t\mathbf{u})}{1 + t^2|\mathbf{u}|^2} \right|_{t=0} = 2\mathbf{u}'\mathbf{A}\mathbf{v}_1 = 0,$$

using symmetry and $\mathbf{v}_1'\mathbf{u} = 0$. So $\mathbf{A}\mathbf{v}_1$ is orthogonal to every $\mathbf{u} \perp \mathbf{v}_1$, which forces $\mathbf{A}\mathbf{v}_1 = \lambda_1\mathbf{v}_1$; and $\lambda_1 = \mathbf{v}_1'\mathbf{A}\mathbf{v}_1$ is real.

The subspace $\mathbf{v}_1^\perp$ is $\mathbf{A}$ -invariant: for $\mathbf{u} \perp \mathbf{v}_1$, $\langle \mathbf{A}\mathbf{u}, \mathbf{v}_1 \rangle = \langle \mathbf{u}, \mathbf{A}\mathbf{v}_1 \rangle = \lambda_1 \langle \mathbf{u}, \mathbf{v}_1 \rangle = 0$. Applying the inductive hypothesis to the restriction of $\mathbf{A}$ to $\mathbf{v}_1^\perp$, a space of dimension $n-1$, yields an orthonormal

eigenbasis there; adjoining $\mathbf{v}_1$ completes it. □

Corollary P.38 (Rayleigh). For symmetric $\mathbf{A}$ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n$, $\max_{|\mathbf{v}|=1} \mathbf{v}'\mathbf{A}\mathbf{v} = \lambda_1$ and $\min_{|\mathbf{v}|=1} \mathbf{v}'\mathbf{A}\mathbf{v} = \lambda_n$.

Proof. Writing $\mathbf{v} = \mathbf{P}\mathbf{w}$ with $|\mathbf{w}| = |\mathbf{v}| = 1$, $\mathbf{v}'\mathbf{A}\mathbf{v} = \sum_j \lambda_j w_j^2$, a convex combination of the eigenvalues, so it lies between λ_n and λ_1 ; the bounds are attained at the corresponding eigenvectors. □

10 Positive Definiteness and Square Roots

Definition P.39. Symmetric $\mathbf{A}$ is **positive semi-definite** if $\mathbf{v}'\mathbf{A}\mathbf{v} \geq 0$ for all $\mathbf{v}$, and **positive definite** if $\mathbf{v}'\mathbf{A}\mathbf{v} > 0$ for $\mathbf{v} \neq \mathbf{0}$.

Proposition P.40. A symmetric $\mathbf{A}$ is positive semi-definite iff all its eigenvalues are non-negative, and positive definite iff all are strictly positive; in the latter case $\mathbf{A}$ is invertible.

Proof. By Corollary P.38 the extreme values of $\mathbf{v}'\mathbf{A}\mathbf{v}$ on the unit sphere are λ_n and λ_1 , so non-negativity of the quadratic form is exactly $\lambda_n \geq 0$ and positivity exactly $\lambda_n > 0$. If every eigenvalue is nonzero then $\mathbf{\Lambda}$, hence $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$, is invertible. $\square$

Theorem P.41 (Symmetric square root). A positive semi-definite $\mathbf{A}$ has a unique positive semi-definite $\mathbf{A}^{1/2}$ with $(\mathbf{A}^{1/2})^2 = \mathbf{A}$. If $\mathbf{A}$ is positive definite so is $\mathbf{A}^{1/2}$, and $\mathbf{A}^{-1/2} := (\mathbf{A}^{1/2})^{-1}$ satisfies $\mathbf{A}^{-1/2}\mathbf{A}\mathbf{A}^{-1/2} = \mathbf{I}$.

Proof. Write $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ and set $\mathbf{A}^{1/2} = \mathbf{P}\mathbf{\Lambda}^{1/2}\mathbf{P}'$ with $\mathbf{\Lambda}^{1/2}$ the diagonal matrix of non-negative square roots, legitimate by Proposition P.40. It is symmetric, positive semi-definite, and squares to $\mathbf{A}$. For uniqueness, let $\mathbf{B}$ be positive semi-definite with $\mathbf{B}^2 = \mathbf{A}$. Any eigenvector $\mathbf{u}$ of $\mathbf{B}$ with eigenvalue $\mu \geq 0$ satisfies $\mathbf{A}\mathbf{u} = \mu^2\mathbf{u}$, so $\mathbf{B}$ and $\mathbf{A}$ share an eigenbasis, and on it $\mathbf{B}$ acts by $\mu = \sqrt{\mu^2}$, the non-negative root, which is how $\mathbf{A}^{1/2}$ acts. $\square$

11 Idempotent Matrices

Definition P.42. $\mathbf{A}$ is **idempotent** if $\mathbf{A}^2 = \mathbf{A}$.

Lemma P.43 (Trace equals rank). An idempotent $\mathbf{A} \in \mathbb{R}^{n \times n}$ has $\text{tr}(\mathbf{A}) = \text{rank}(\mathbf{A})$, and its eigenvalues lie in $\{0, 1\}$.

Proof. If $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$ then $\lambda\mathbf{v} = \mathbf{A}^2\mathbf{v} = \lambda^2\mathbf{v}$, so $\lambda \in \{0, 1\}$. Next, $\mathbb{R}^n = \mathcal{N}(\mathbf{A}) \oplus \mathcal{C}(\mathbf{A})$: any $\mathbf{v}$ splits as $(\mathbf{v} - \mathbf{A}\mathbf{v}) + \mathbf{A}\mathbf{v}$ with $\mathbf{A}(\mathbf{v} - \mathbf{A}\mathbf{v}) = \mathbf{0}$, and if $\mathbf{v} = \mathbf{A}\mathbf{w}$ lies in both then $\mathbf{v} = \mathbf{A}^2\mathbf{w} = \mathbf{A}\mathbf{v} = \mathbf{0}$. Every $\mathbf{u} \in \mathcal{C}(\mathbf{A})$ has $\mathbf{A}\mathbf{u} = \mathbf{u}$, so a basis of $\mathcal{N}(\mathbf{A})$ together with one of $\mathcal{C}(\mathbf{A})$ is an eigenbasis and $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}$ with $r = \text{rank}(\mathbf{A})$ ones and $n - r$ zeros on the diagonal. The trace is invariant under similarity, so $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{\Lambda}) = r$. $\square$

Proposition P.44 (Projections are the symmetric idempotents). $\mathbf{H}$ is the matrix of the orthogonal projection onto a subspace S if and only if $\mathbf{H}$ is symmetric, idempotent, and $\mathcal{C}(\mathbf{H}) = S$.

Proof. For $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ symmetry and $\mathbf{H}^2 = \mathbf{H}$ are direct computations, and $\mathbf{H}\mathbf{X} = \mathbf{X}$ gives $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$. Conversely let $\mathbf{H}$ be symmetric idempotent with column space S . For $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{H}\mathbf{y} \in S$ and $\mathbf{y} - \mathbf{H}\mathbf{y} \perp S$, because for $\mathbf{u} = \mathbf{H}\mathbf{w} \in S$, $\langle \mathbf{y} - \mathbf{H}\mathbf{y}, \mathbf{H}\mathbf{w} \rangle = \mathbf{w}'\mathbf{H}'(\mathbf{I} - \mathbf{H})\mathbf{y} = \mathbf{w}'(\mathbf{H} - \mathbf{H}^2)\mathbf{y} = 0$. By the characterisation in Theorem P.35, $\mathbf{H}\mathbf{y}$ is the projection. $\square$

12 Block Matrices

Theorem P.45 (Schur complement). Let $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$ with $\mathbf{A}$ invertible, and put $\mathbf{S} = \mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}$. If $\mathbf{S}$ is invertible then $\mathbf{M}$ is, with

$$\mathbf{M}^{-1} = \begin{pmatrix} \mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} & -\mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1} \\ -\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} & \mathbf{S}^{-1} \end{pmatrix}, \quad (\text{P.11})$$

so the lower-right block of $\mathbf{M}^{-1}$ is $\mathbf{S}^{-1}$.

Proof. Verify $\mathbf{M}\mathbf{M}^{-1} = \mathbf{I}$ blockwise. The (1, 1) block is $\mathbf{A}(\mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1}) - \mathbf{B}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} = \mathbf{I} + \mathbf{B}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} - \mathbf{B}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} = \mathbf{I}$; the (1, 2) block is $-\mathbf{A}\mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1} + \mathbf{B}\mathbf{S}^{-1} = \mathbf{0}$; the (2, 1) block

is $\mathbf{CA}^{-1} + \mathbf{CA}^{-1}\mathbf{BS}^{-1}\mathbf{CA}^{-1} - \mathbf{DS}^{-1}\mathbf{CA}^{-1} = \mathbf{CA}^{-1} - \mathbf{SS}^{-1}\mathbf{CA}^{-1} = \mathbf{0}$; and the $(2,2)$ block is $-\mathbf{CA}^{-1}\mathbf{BS}^{-1} + \mathbf{DS}^{-1} = \mathbf{SS}^{-1} = \mathbf{I}$. $\square$

Corollary P.46. For $\mathbf{W} = [\mathbf{1} \ \mathbf{Z}]$ with n rows, the lower-right $p \times p$ block of $(\mathbf{W}'\mathbf{W})^{-1}$ is $(\mathbf{Z}'\mathbf{M}_0\mathbf{Z})^{-1}$, where $\mathbf{M}_0 = \mathbf{I}_n - n^{-1}\mathbf{1}\mathbf{1}'$.

Proof. $\mathbf{W}'\mathbf{W} = \begin{pmatrix} n & \mathbf{1}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{1} & \mathbf{Z}'\mathbf{Z} \end{pmatrix}$, whose Schur complement of the $(1,1)$ entry is $\mathbf{Z}'\mathbf{Z} - \mathbf{Z}'\mathbf{1}n^{-1}\mathbf{1}'\mathbf{Z} = \mathbf{Z}'\mathbf{M}_0\mathbf{Z}$. Apply Theorem P.45. $\square$

Lemma P.47 (Sherman–Morrison). Let $\mathbf{A}$ be invertible and $\mathbf{u}, \mathbf{v}$ vectors with $1 - \mathbf{v}'\mathbf{A}^{-1}\mathbf{u} \neq 0$. Then

$$(\mathbf{A} - \mathbf{u}\mathbf{v}')^{-1} = \mathbf{A}^{-1} + \frac{\mathbf{A}^{-1}\mathbf{u}\mathbf{v}'\mathbf{A}^{-1}}{1 - \mathbf{v}'\mathbf{A}^{-1}\mathbf{u}}. \quad (\text{P.12})$$

Proof. Write $c = 1 - \mathbf{v}'\mathbf{A}^{-1}\mathbf{u}$ and multiply the right side by $\mathbf{A} - \mathbf{u}\mathbf{v}'$:

$$\mathbf{I} - \mathbf{A}^{-1}\mathbf{u}\mathbf{v}' + \frac{\mathbf{A}^{-1}\mathbf{u}\mathbf{v}' - \mathbf{A}^{-1}\mathbf{u}(\mathbf{v}'\mathbf{A}^{-1}\mathbf{u})\mathbf{v}'}{c} = \mathbf{I} - \mathbf{A}^{-1}\mathbf{u}\mathbf{v}' + \frac{\mathbf{A}^{-1}\mathbf{u}\mathbf{v}'c}{c} = \mathbf{I},$$

using that $\mathbf{v}'\mathbf{A}^{-1}\mathbf{u}$ is a scalar. □

13 The Singular Value Decomposition

Theorem P.48 (SVD). Every $\mathbf{A} \in \mathbb{R}^{n \times p}$ factors as $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}'$ with $\mathbf{U} \in \mathbb{R}^{n \times r}$ and $\mathbf{V} \in \mathbb{R}^{p \times r}$ having orthonormal columns, $r = \text{rank}(\mathbf{A})$, and $\mathbf{D} = \text{diag}(d_1, \dots, d_r)$ with $d_1 \geq \dots \geq d_r > 0$. The d_j^2 are the nonzero eigenvalues of $\mathbf{A}'\mathbf{A}$.

Proof. $\mathbf{A}'\mathbf{A}$ is symmetric positive semi-definite, so by Theorem P.37 and Proposition P.40 it has an orthonormal eigenbasis $\mathbf{v}_1, \dots, \mathbf{v}_p$ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_p \geq 0$. Since $|\mathbf{A}\mathbf{v}_j|^2 = \lambda_j$, exactly $r = \text{rank}(\mathbf{A}'\mathbf{A}) = \text{rank}(\mathbf{A})$ of them are positive by Proposition P.31. Put $d_j = \sqrt{\lambda_j}$ and $\mathbf{u}_j = \mathbf{A}\mathbf{v}_j/d_j$ for $j \leq r$; these are orthonormal, since $\mathbf{u}_j'\mathbf{u}_k = \mathbf{v}_j'\mathbf{A}'\mathbf{A}\mathbf{v}_k/(d_j d_k) = \lambda_k \delta_{jk}/(d_j d_k)$. For any $\mathbf{x}$, expand $\mathbf{x} = \sum_{j \leq p} \langle \mathbf{v}_j, \mathbf{x} \rangle \mathbf{v}_j$. The terms with $j > r$ have $\mathbf{A}\mathbf{v}_j = \mathbf{0}$, so $\mathbf{A}\mathbf{x} = \sum_{j \leq r} \langle \mathbf{v}_j, \mathbf{x} \rangle d_j \mathbf{u}_j = \mathbf{U}\mathbf{D}\mathbf{V}'\mathbf{x}$. □

14 Norms and Conditioning

Definition P.49 (Operator norm, condition number). $\|\mathbf{A}\| = \sup_{|\mathbf{x}|=1} |\mathbf{Ax}|$. For invertible $\mathbf{A}$ the **condition number** is $\kappa(\mathbf{A}) = \|\mathbf{A}\| \|\mathbf{A}^{-1}\|$.

Proposition P.50. $\|\mathbf{A}\| = d_1$, the largest singular value of $\mathbf{A}$. For symmetric positive definite $\mathbf{A}$ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n > 0$, $\|\mathbf{A}\| = \lambda_1$, $\|\mathbf{A}^{-1}\| = \lambda_n^{-1}$ and $\kappa(\mathbf{A}) = \lambda_1/\lambda_n$.

Proof. $|\mathbf{Ax}|^2 = \mathbf{x}'\mathbf{A}'\mathbf{Ax}$, and by Corollary P.38 its maximum over $|\mathbf{x}| = 1$ is the largest eigenvalue of $\mathbf{A}'\mathbf{A}$, which is d_1^2 by Theorem P.48. For symmetric positive definite $\mathbf{A}$ the singular values are the eigenvalues, since $\mathbf{A}'\mathbf{A} = \mathbf{A}^2$ has eigenvalues λ_i^2 ; and $\mathbf{A}^{-1}$ has eigenvalues λ_i^{-1} , the largest being λ_n^{-1} . $\square$

Theorem P.51 (Conditioning of a linear system). Let $\mathbf{A}$ be invertible, $\mathbf{Ax} = \mathbf{b} \neq \mathbf{0}$, and $\mathbf{A}(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b} + \delta\mathbf{b}$. Then

$$\frac{|\delta\mathbf{x}|}{|\mathbf{x}|} \leq \kappa(\mathbf{A}) \frac{|\delta\mathbf{b}|}{|\mathbf{b}|}. \quad (\text{P.13})$$

Proof. Subtracting the two systems gives $\mathbf{A} \delta \mathbf{x} = \delta \mathbf{b}$, so $|\delta \mathbf{x}| = |\mathbf{A}^{-1} \delta \mathbf{b}| \leq \|\mathbf{A}^{-1}\| |\delta \mathbf{b}|$. Also $|\mathbf{b}| = |\mathbf{A} \mathbf{x}| \leq \|\mathbf{A}\| |\mathbf{x}|$, i.e. $1/|\mathbf{x}| \leq \|\mathbf{A}\|/|\mathbf{b}|$. Multiplying the two bounds gives (P.13). $\square$

Corollary P.52 (Why the normal equations square the difficulty). For $\mathbf{X}$ of full column rank, $\kappa(\mathbf{X}'\mathbf{X}) = \kappa(\mathbf{X})^2$, where $\kappa(\mathbf{X})$ is the ratio of the largest to the smallest singular value of $\mathbf{X}$.

Proof. By Theorem P.48, $\mathbf{X}'\mathbf{X} = \mathbf{V}\mathbf{D}^2\mathbf{V}'$ has eigenvalues d_j^2 , so by Proposition P.50 its condition number is $d_1^2/d_r^2 = (d_1/d_r)^2$. $\square$

Remark. Corollary P.52 is the reason least squares is computed from a **QR** factorisation of $\mathbf{X}$ rather than by forming $\mathbf{X}'\mathbf{X}$: the latter squares the condition number, so a design that is merely ill-conditioned becomes numerically singular. It is also the quantitative form of the collinearity diagnostics of Unit 4: a large $\kappa(\mathbf{X}'\mathbf{X})$ and a large variance inflation factor detect the same near-degeneracy, the first through the spectrum of $\mathbf{X}'\mathbf{X}$ and the second one predictor at a time.

15 Matrix Differentiation

Definition P.53 (Differential and derivative). For $f : \mathbb{R}^p \rightarrow \mathbb{R}$ differentiable at $\boldsymbol{\beta}$, the **differential** is the linear form $df = \sum_j (\partial f / \partial b_j) db_j$, and $Df(\boldsymbol{\beta}) := \partial f / \partial \boldsymbol{\beta}'$ is the row vector of partial derivatives.

Proposition P.54 (First identification theorem). If $df(\boldsymbol{\beta}) = \mathbf{c}(\boldsymbol{\beta})' d\boldsymbol{\beta}$ for a vector $\mathbf{c}(\boldsymbol{\beta})$, then $Df(\boldsymbol{\beta}) = \mathbf{c}(\boldsymbol{\beta})'$.

Proof. Both sides are the coefficient vector of the same linear form in $d\boldsymbol{\beta}$, and a linear form determines its coefficients: apply it to the standard basis vectors. $\square$

Lemma P.55 (The two rules used in the course). For constant $\mathbf{a} \in \mathbb{R}^p$ and constant $\mathbf{A} \in \mathbb{R}^{p \times p}$,

- (i) $D(\mathbf{a}'\boldsymbol{\beta}) = D(\boldsymbol{\beta}'\mathbf{a}) = \mathbf{a}'$;
- (ii) $D(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}) = \boldsymbol{\beta}'(\mathbf{A} + \mathbf{A}')$, which is $2\boldsymbol{\beta}'\mathbf{A}$ when $\mathbf{A}$ is symmetric.

Proof. (i) $d(\mathbf{a}'\boldsymbol{\beta}) = \mathbf{a}'d\boldsymbol{\beta}$ since $d\mathbf{a} = 0$; apply Proposition P.54, and note $\boldsymbol{\beta}'\mathbf{a} = \mathbf{a}'\boldsymbol{\beta}$.

(ii) With $d\mathbf{A} = 0$, the product rule and the scalar identity $(d\boldsymbol{\beta})'\mathbf{A}\boldsymbol{\beta} = \boldsymbol{\beta}'\mathbf{A}'d\boldsymbol{\beta}$ give $d(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}) = (d\boldsymbol{\beta})'\mathbf{A}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{A}d\boldsymbol{\beta} = \boldsymbol{\beta}'(\mathbf{A} + \mathbf{A}')d\boldsymbol{\beta}$. $\square$

Proposition P.56 (Chain rule for differentials). Let $\mathbf{u} = g(\mathbf{x})$ be differentiable at $\mathbf{x}$ and f differentiable at $\mathbf{u}$. Then the differential of the composition is obtained by substitution: $df = (Df)(\mathbf{u})d\mathbf{u}$ with $d\mathbf{u} = (Dg)(\mathbf{x})d\mathbf{x}$, hence $D(f \circ g)(\mathbf{x}) = (Df)(g(\mathbf{x}))(Dg)(\mathbf{x})$.

Proof. Differentiability of f at $\mathbf{u}$ gives $f(\mathbf{u} + \mathbf{h}) - f(\mathbf{u}) = (Df)(\mathbf{u})\mathbf{h} + o(|\mathbf{h}|)$, and of g at $\mathbf{x}$ gives $g(\mathbf{x} + \mathbf{k}) - g(\mathbf{x}) = (Dg)(\mathbf{x})\mathbf{k} + o(|\mathbf{k}|)$. Substituting the second into the first with $\mathbf{h} = g(\mathbf{x} + \mathbf{k}) - g(\mathbf{x})$, and using $|\mathbf{h}| = O(|\mathbf{k}|)$,

$$f(g(\mathbf{x} + \mathbf{k})) - f(g(\mathbf{x})) = (Df)(\mathbf{u})(Dg)(\mathbf{x})\mathbf{k} + o(|\mathbf{k}|),$$

which by Proposition P.54 identifies the derivative. $\square$

Example P.1 (The form used for likelihoods). For a scalar h and the linear index $\eta = \mathbf{x}'\boldsymbol{\beta}$,

$$\frac{\partial}{\partial \boldsymbol{\beta}'} h(\mathbf{x}'\boldsymbol{\beta}) = h'(\mathbf{x}'\boldsymbol{\beta}) \mathbf{x}',$$

since $d\eta = \mathbf{x}'d\boldsymbol{\beta}$ by Lemma P.55(i) and $dh = h'(\eta) d\eta$. Summing over observations gives the score vector of a generalised linear model, and differentiating once more with Proposition P.56 gives the information matrix.

Part III

Measure-Theoretic Probability

16 The Integral and Its Convergence Theorems

A **measure space** is a triple $(\Omega, \mathcal{F}, \mu)$ with $\mathcal{F}$ a σ -field and μ a countably additive non-negative set function with $\mu(\emptyset) = 0$; it is a **probability space**, written $(\Omega, \mathcal{F}, \mathbf{P})$, when $\mathbf{P}(\Omega) = 1$. A **random vector** is a measurable $X : \Omega \rightarrow \mathbb{R}^d$, its **law** is the induced measure $\mu_X(B) = \mathbf{P}\{X \in B\}$, and $\mathbf{E}[g(X)] = \int g \, d\mu_X$ when the integral exists.

Remark (The four assumed theorems). The following are the only statements in the course used without proof.

- (i) **Monotone convergence.** If $0 \leq f_n \uparrow f$ then $\int f_n \uparrow \int f$.
- (ii) **Fatou.** If $f_n \geq 0$ then $\int \liminf_n f_n \leq \liminf_n \int f_n$.

- (iii) **Dominated convergence.** If $f_n \rightarrow f$ pointwise and $|f_n| \leq g$ with $\int g < \infty$, then $\int f_n \rightarrow \int f$.
- (iv) **Fubini–Tonelli.** For σ -finite measures, a non-negative or absolutely integrable function of two variables may be integrated in either order.

Everything else in this document is derived from them.

17 Expectation and Inequalities

Theorem P.57 (Markov, Chebyshev). For a random variable $X \geq 0$ and $a > 0$, $\mathbf{P}\{X \geq a\} \leq \mathbf{E}[X]/a$. Consequently $\mathbf{P}\{|Y - \mathbf{E}Y| \geq a\} \leq \text{var}(Y)/a^2$.

Proof. $a \mathbf{1}\{X \geq a\} \leq X$ pointwise; take expectations. For the second, apply the first to $X = (Y - \mathbf{E}Y)^2$ and the level a^2 . $\square$

Theorem P.58 (Jensen). If ϕ is convex on an open interval containing the range of an integrable X , then $\phi(\mathbf{E}X) \leq \mathbf{E}[\phi(X)]$.

Proof. Convexity gives a supporting line at $m = \mathbf{E} X$: by Proposition P.21 in the differentiable case, and in general by taking c between the one-sided derivatives, $\phi(x) \geq \phi(m) + c(x - m)$ for all x . Substituting X and taking expectations kills the linear term. $\square$

Proposition P.59 (Variance under a linear map). If $\mathbf{z}$ has covariance $\text{var}(\mathbf{z})$, and $\mathbf{A}$, $\mathbf{b}$ are constant, then $\text{var}(\mathbf{A}\mathbf{z} + \mathbf{b}) = \mathbf{A} \text{var}(\mathbf{z}) \mathbf{A}'$.

Proof. With $\boldsymbol{\mu} = \mathbf{E}[\mathbf{z}]$, $\mathbf{E}[\mathbf{A}\mathbf{z} + \mathbf{b}] = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$, so

$$\text{var}(\mathbf{A}\mathbf{z} + \mathbf{b}) = \mathbf{E}[\mathbf{A}(\mathbf{z} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\mu})' \mathbf{A}'] = \mathbf{A} \mathbf{E}[(\mathbf{z} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\mu})'] \mathbf{A}'. \quad \square$$

Lemma P.60 (Expectation of a quadratic form). If $\mathbf{E}[\mathbf{z}] = \boldsymbol{\mu}$ and $\text{var}(\mathbf{z}) = \boldsymbol{\Sigma}$, and $\mathbf{A}$ is a constant square matrix, then $\mathbf{E}[\mathbf{z}' \mathbf{A} \mathbf{z}] = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}' \mathbf{A} \boldsymbol{\mu}$.

Proof. $\mathbf{E}[\mathbf{z}' \mathbf{A} \mathbf{z}] = \sum_{i,j} a_{ij} \mathbf{E}[z_i z_j]$ and $\mathbf{E}[z_i z_j] = \Sigma_{ij} + \mu_i \mu_j$. The first part is $\sum_{i,j} a_{ij} \Sigma_{ij} = \text{tr}(\mathbf{A} \boldsymbol{\Sigma})$ by symmetry of $\boldsymbol{\Sigma}$, and the second is $\boldsymbol{\mu}' \mathbf{A} \boldsymbol{\mu}$. $\square$

18 Independence and Borel–Cantelli

Lemma P.61 (Borel–Cantelli). If $\sum_k \mathbf{P}(A_k) < \infty$ then $\mathbf{P}\{A_k \text{ infinitely often}\} = 0$.

Proof. Let $N = \sum_k \mathbf{1}_{A_k}$. By monotone convergence $\mathbf{E}[N] = \sum_k \mathbf{P}(A_k) < \infty$, so $N < \infty$ almost surely. $\square$

19 Conditional Expectation

Unit 1 defines the regression function as $f^*(x) = \mathbf{E}[Y \mid X = x]$ and proves it is the best squared-error predictor. That statement rests on the following construction, which is itself an orthogonal projection, the same one as Theorem P.35, in L^2 rather than $\mathbb{R}^n$.

Definition P.62 (L^2 and its inner product). $L^2(\Omega, \mathcal{F}, \mathbf{P})$ is the space of random variables with $\mathbf{E}[X^2] < \infty$, with $\langle X, Y \rangle = \mathbf{E}[XY]$, variables equal almost surely being identified.

Theorem P.63 (Conditional expectation in L^2). Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub- σ -field and $X \in L^2(\mathcal{F})$.

There is a $\mathcal{G}$ -measurable $\mathbf{E}[X \mid \mathcal{G}] \in L^2(\mathcal{G})$, unique up to null sets, with

$$\mathbf{E}[(X - \mathbf{E}[X \mid \mathcal{G}]) W] = 0 \quad \text{for every } W \in L^2(\mathcal{G}), \quad (\text{P.14})$$

and it minimises $\mathbf{E}[(X - W)^2]$ over $W \in L^2(\mathcal{G})$.

Proof. $L^2(\mathcal{G})$ is a closed convex subset of the complete space $L^2(\mathcal{F})$: closed because an L^2 -convergent sequence has an almost surely convergent subsequence, whose limit is $\mathcal{G}$ -measurable. Theorem P.19, whose proof used only the parallelogram law and completeness and so applies verbatim in L^2 , gives a unique nearest point $\hat{X}$ and the variational inequality $\langle X - \hat{X}, W - \hat{X} \rangle \leq 0$ for all $W \in L^2(\mathcal{G})$. Since $L^2(\mathcal{G})$ is a subspace the inequality holds for W and for $2\hat{X} - W$, forcing equality, which is (P.14). Minimality is Theorem P.35's characterisation transported to L^2 . $\square$

Remark. Taking $W = \mathbf{1}_G$ for $G \in \mathcal{G}$ turns (P.14) into the familiar $\mathbf{E}[X \mathbf{1}_G] = \mathbf{E}[\mathbf{E}[X \mid \mathcal{G}] \mathbf{1}_G]$. For $X \in L^1 \setminus L^2$ the same object is obtained by approximating with truncations $X \wedge m \vee (-m)$ and passing to the limit by dominated convergence; the course needs only the L^2 case, where the projection interpretation is exactly what Unit 1 uses.

Proposition P.64 (Tower property). $\mathbf{E}[\mathbf{E}[X \mid \mathcal{G}]] = \mathbf{E}[X]$.

Proof. Take $W \equiv 1$ in (P.14). □

Proposition P.65 (Conditional variance decomposition). For $X \in L^2$, writing $\text{var}(X \mid \mathcal{G}) = \mathbf{E}[X^2 \mid \mathcal{G}] - (\mathbf{E}[X \mid \mathcal{G}])^2$,

$$\text{var}(X) = \mathbf{E}[\text{var}(X \mid \mathcal{G})] + \text{var}(\mathbf{E}[X \mid \mathcal{G}]). \quad (\text{P.15})$$

Proof. By Proposition P.64 applied to X^2 and to X ,

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E} X)^2 = \mathbf{E}[\mathbf{E}[X^2 \mid \mathcal{G}]] - (\mathbf{E}[\mathbf{E}[X \mid \mathcal{G}]])^2.$$

Substituting $\mathbf{E}[X^2 \mid \mathcal{G}] = \text{var}(X \mid \mathcal{G}) + (\mathbf{E}[X \mid \mathcal{G}])^2$ and recognising $\mathbf{E}[(\mathbf{E}[X \mid \mathcal{G}])^2] - (\mathbf{E}[\mathbf{E}[X \mid \mathcal{G}]])^2$ as $\text{var}(\mathbf{E}[X \mid \mathcal{G}])$ gives (P.15). □

Corollary P.66 (Random sums). Let N be a non-negative integer random variable and $Y_1, Y_2, \dots$ independent copies of a square-integrable Y , independent of N , and $S = \sum_{j \leq N} Y_j$. Then

$$\mathbf{E}[S] = \mathbf{E}[N] \mathbf{E}[Y], \quad \text{var}(S) = \mathbf{E}[N] \text{var}(Y) + \text{var}(N)(\mathbf{E} Y)^2.$$

Proof. Conditionally on $N = m$ the sum is of m independent copies, so $\mathbf{E}[S \mid N] = N \mathbf{E}[Y]$ and $\text{var}(S \mid N) = N \text{var}(Y)$. Proposition P.64 gives the mean and Proposition P.65 gives $\text{var}(S) = \mathbf{E}[N \text{var}(Y)] + \text{var}(N \mathbf{E}[Y])$, which is the stated expression. $\square$

20 The Multivariate Normal Distribution

Definition P.67 (Multivariate normal). X in $\mathbb{R}^d$ is $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ if $X = \boldsymbol{\mu} + \mathbf{A}Z$ where Z has independent standard normal coordinates and $\mathbf{A}\mathbf{A}' = \boldsymbol{\Sigma}$.

Theorem P.68 (Closure under affine maps). If $X \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\mathbf{B}$ is a constant matrix and $\mathbf{b}$ a constant vector, then $\mathbf{B}X + \mathbf{b} \sim N(\mathbf{B}\boldsymbol{\mu} + \mathbf{b}, \mathbf{B}\boldsymbol{\Sigma}\mathbf{B}')$.

Proof. $\mathbf{B}X + \mathbf{b} = (\mathbf{B}\boldsymbol{\mu} + \mathbf{b}) + (\mathbf{B}\mathbf{A})Z$ and $(\mathbf{B}\mathbf{A})(\mathbf{B}\mathbf{A})' = \mathbf{B}\boldsymbol{\Sigma}\mathbf{B}'$, which is Definition P.67. $\square$

Theorem P.69 (Uncorrelated blocks are independent). Let $(\mathbf{U}', \mathbf{V}')$ be jointly normal with cross-covariance Σ_{UV} . Then $\Sigma_{UV} = \mathbf{0}$ if and only if $\mathbf{U}$ and $\mathbf{V}$ are independent.

Proof. Independence always gives zero covariance. Conversely, with $\Sigma_{UV} = \mathbf{0}$ the covariance is block diagonal, so by Proposition P.82 below the joint characteristic function factors,

$$\exp(i\langle \mathbf{s}, \boldsymbol{\mu}_U \rangle + i\langle \mathbf{t}, \boldsymbol{\mu}_V \rangle - \frac{1}{2}\mathbf{s}'\Sigma_{UU}\mathbf{s} - \frac{1}{2}\mathbf{t}'\Sigma_{VV}\mathbf{t}) = \varphi_{\mathbf{U}}(\mathbf{s}) \varphi_{\mathbf{V}}(\mathbf{t}),$$

and a joint characteristic function that factors identifies the joint law as the product of the marginals, by the uniqueness Corollary P.88. $\square$

21 The χ^2 , t and F Distributions

Definition P.70. For independent $Z_1, \dots, Z_r$ standard normal, $\sum_{j \leq r} Z_j^2$ has the **chi-square** law with r degrees of freedom, χ_r^2 . If $Z \sim N(0, 1)$ and $V \sim \chi_\nu^2$ are independent then $Z/\sqrt{V/\nu}$ has **Student's** t law t_ν . If $V_1 \sim \chi_{r_1}^2$ and $V_2 \sim \chi_{r_2}^2$ are independent then $(V_1/r_1)/(V_2/r_2)$ has the F law F_{r_1, r_2} .

Corollary P.71 (The square of a t is an F). If $T \sim t_\nu$ then $T^2 \sim F_{1,\nu}$.

Proof. By Definition P.70, $T = Z/\sqrt{V/\nu}$ with $Z \sim N(0, 1)$ and $V \sim \chi_\nu^2$ independent. Squaring, $T^2 = (Z^2/1)/(V/\nu)$. Now $Z^2 \sim \chi_1^2$ by the same definition with $r = 1$, and it is independent of V because Z is, so the ratio is $F_{1,\nu}$. $\square$

Proposition P.72 (Moments of χ_r^2). If $V \sim \chi_r^2$ then $\mathbf{E}[V] = r$ and $\text{var}(V) = 2r$.

Proof. $\mathbf{E}[Z^2] = 1$ and, by Lemma P.73 below, $\mathbf{E}[Z^4] = 3$, so $\text{var}(Z^2) = 3 - 1 = 2$. Sum over r independent terms. $\square$

Lemma P.73 (Fourth moment of the normal). If $Z \sim N(0, 1)$ then $\mathbf{E}[Z^4] = 3$, so $\text{var}(Z^2) = 2$; for $\varepsilon \sim N(0, \sigma^2)$, $\text{var}(\varepsilon^2/\sigma^2) = 2$.

Proof. With $\varphi'(z) = -z\varphi(z)$ for the standard normal density, integration by parts gives

$$\mathbf{E}[Z^4] = \int z^3 \cdot z\varphi(z) \, dz = - \int z^3 \varphi'(z) \, dz = [-z^3 \varphi(z)]_{-\infty}^{\infty} + 3 \int z^2 \varphi(z) \, dz = 3,$$

the boundary term vanishing because φ decays faster than any polynomial grows. $\square$

Theorem P.74 (Chi-square quadratic forms). If $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$ and $\mathbf{A}$ is symmetric idempotent, then $\mathbf{z}'\mathbf{A}\mathbf{z} \sim \chi_r^2$ with $r = \text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A})$.

Proof. By Theorem P.37 write $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ with $\mathbf{P}$ orthogonal; by Lemma P.43 the diagonal of $\mathbf{\Lambda}$ carries r ones and $n - r$ zeros. Put $\mathbf{w} = \mathbf{P}'\mathbf{z}$, which is $N(\mathbf{0}, \mathbf{P}'\mathbf{I}\mathbf{P}) = N(\mathbf{0}, \mathbf{I}_n)$ by Theorem P.68, so its coordinates are independent standard normals. Then $\mathbf{z}'\mathbf{A}\mathbf{z} = \mathbf{w}'\mathbf{\Lambda}\mathbf{w} = \sum_{j \leq r} w_j^2$, which is χ_r^2 by Definition P.70. $\square$

Theorem P.75 (Cochran). If $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$ and $\mathbf{A}_1, \mathbf{A}_2$ are symmetric idempotent with $\mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$, then $\mathbf{z}'\mathbf{A}_1\mathbf{z}$ and $\mathbf{z}'\mathbf{A}_2\mathbf{z}$ are independent.

Proof. Stack $\mathbf{U} = \mathbf{A}_1\mathbf{z}$ and $\mathbf{V} = \mathbf{A}_2\mathbf{z}$ as $(\mathbf{U}', \mathbf{V}')' = (\mathbf{A}'_1, \mathbf{A}'_2)'\mathbf{z}$, one linear map of one normal vector, hence jointly normal by Theorem P.68; individual normality would not suffice. The cross-covariance is $\mathbf{A}_1\mathbf{I}\mathbf{A}'_2 = \mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$, so Theorem P.69 makes $\mathbf{U}$ and $\mathbf{V}$ independent, and $\mathbf{z}'\mathbf{A}_i\mathbf{z} = |\mathbf{A}_i\mathbf{z}|^2$ is a function of one of them alone. $\square$

22 Modes of Convergence

$C_b(\mathbb{R}^d)$ denotes the bounded continuous real functions on $\mathbb{R}^d$ and $C_K(\mathbb{R}^d)$ those continuous with compact support.

Definition P.76 (Modes of convergence). Let $X, X_1, X_2, \dots$ be random vectors in $\mathbb{R}^d$.

- (i) $X_n \rightarrow X$ **almost surely**, written $X_n \rightarrow X$ a.s., if $\mathbf{P}\{\lim_n X_n = X\} = 1$.
- (ii) $X_n \xrightarrow{p} X$ **in probability** if $\mathbf{P}\{|X_n - X| > \varepsilon\} \rightarrow 0$ for every $\varepsilon > 0$.
- (iii) $X_n \xrightarrow{d} X$ **in distribution** if $\mathbf{E}[f(X_n)] \rightarrow \mathbf{E}[f(X)]$ for every $f \in C_b(\mathbb{R}^d)$.

Convergence in distribution depends on the X_n only through their laws, so we also write $\mu_n \rightarrow \mu$ **weakly** for probability measures.

Lemma P.77 (Almost sure subsequence). If $X_n \xrightarrow{p} X$ then some subsequence converges to X almost surely.

Proof. Choose $n_1 < n_2 < \dots$ with $\mathbf{P}\{|X_{n_k} - X| > 2^{-k}\} \leq 2^{-k}$, possible because each probability tends to 0. These probabilities are summable, so by Lemma P.61 the events $\{|X_{n_k} - X| > 2^{-k}\}$

occur finitely often almost surely; on that event $|X_{n_k} - X| \leq 2^{-k}$ for all large k , hence $X_{n_k} \rightarrow X$. $\square$

Proposition P.78 (The hierarchy). Almost sure convergence implies convergence in probability, which implies convergence in distribution.

Proof. a.s. $\Rightarrow$ probability. Fix $\varepsilon > 0$ and put $B_n = \{|X_n - X| > \varepsilon\}$. On the event $\{X_n \rightarrow X\}$ only finitely many B_n occur, so $\mathbb{1}_{B_n} \rightarrow 0$ almost surely; dominated convergence gives $\mathbb{P}(B_n) = \mathbb{E}[\mathbb{1}_{B_n}] \rightarrow 0$.

probability $\Rightarrow$ distribution. Let $f \in C_b(\mathbb{R}^d)$ and suppose $\mathbb{E}[f(X_n)] \not\rightarrow \mathbb{E}[f(X)]$. Then some subsequence satisfies $|\mathbb{E}[f(X_{n_j})] - \mathbb{E}[f(X)]| \geq \delta$ for a $\delta > 0$. That subsequence still converges in probability, so by Lemma P.77 it has a further subsequence converging almost surely; along it $f(X_{n_{j_l}}) \rightarrow f(X)$ almost surely by continuity, and dominated convergence (with dominating constant $\sup |f|$) gives $\mathbb{E}[f(X_{n_{j_l}})] \rightarrow \mathbb{E}[f(X)]$, contradicting the choice of δ . $\square$

Proposition P.79 (Convergence to a constant). If $c \in \mathbb{R}^d$ is constant then $X_n \xrightarrow{d} c$ if and only if $X_n \xrightarrow{p} c$.

Proof. One direction is Proposition P.78. Conversely suppose $X_n \xrightarrow{d} c$ and fix $\varepsilon > 0$. The function $f(x) = \min\{1, |x - c|/\varepsilon\}$ is bounded and continuous with $f(c) = 0$, and $\mathbf{1}_{\{|x - c| > \varepsilon\}} \leq f(x)$ for every x . Hence

$$\mathbf{P}\{|X_n - c| > \varepsilon\} \leq \mathbf{E}[f(X_n)] \rightarrow \mathbf{E}[f(c)] = 0. \quad \square$$

23 Characteristic Functions

Definition P.80 (Characteristic function). The **characteristic function** of a random vector X in $\mathbb{R}^d$, equivalently of its law μ , is

$$\varphi_X(\theta) = \mathbf{E}[e^{i\langle \theta, X \rangle}] = \int_{\mathbb{R}^d} e^{i\langle \theta, x \rangle} d\mu(x), \quad \theta \in \mathbb{R}^d. \quad (\text{P.16})$$

Proposition P.81 (Elementary properties).

- (i) $\varphi_X(0) = 1$ and $|\varphi_X(\theta)| \leq 1$ for all θ .
- (ii) φ_X is uniformly continuous.
- (iii) If X and Y are independent then $\varphi_{X+Y} = \varphi_X \varphi_Y$.
- (iv) For a constant matrix $\mathbf{A}$ and vector $\mathbf{b}$, $\varphi_{\mathbf{A}X+\mathbf{b}}(\theta) = e^{i\langle\theta, \mathbf{b}\rangle} \varphi_X(\mathbf{A}'\theta)$.

Proof. (i) is immediate from $|e^{i\langle\theta, x\rangle}| = 1$. For (ii),

$$|\varphi_X(\theta + \eta) - \varphi_X(\theta)| = |\mathbf{E}[e^{i\langle\theta, X\rangle}(e^{i\langle\eta, X\rangle} - 1)]| \leq \mathbf{E}|e^{i\langle\eta, X\rangle} - 1|,$$

a bound free of θ ; the integrand is at most 2 and tends to 0 pointwise as $\eta \rightarrow 0$, so dominated convergence makes it vanish. (iii) is $\mathbf{E}[e^{i\langle\theta, X\rangle} e^{i\langle\theta, Y\rangle}] = \mathbf{E}[e^{i\langle\theta, X\rangle}] \mathbf{E}[e^{i\langle\theta, Y\rangle}]$ by independence, and (iv) follows from $\langle\theta, \mathbf{A}X + \mathbf{b}\rangle = \langle\mathbf{A}'\theta, X\rangle + \langle\theta, \mathbf{b}\rangle$. $\square$

Proposition P.82 (The Gaussian characteristic function). If $X \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ in $\mathbb{R}^d$ then $\varphi_X(\theta) = \exp(i\langle\theta, \boldsymbol{\mu}\rangle - \frac{1}{2}\theta'\boldsymbol{\Sigma}\theta)$.

Proof. Take first $d = 1$, $Z \sim N(0, 1)$, and set $\varphi(t) = \mathbf{E}[e^{itZ}]$. Differentiating under the integral sign, justified because $|\partial_t e^{itz}| = |z|$ is integrable against the standard normal density ϕ ,

$$\varphi'(t) = \mathbf{E}[iZe^{itZ}] = i \int ze^{itz} \phi(z) dz = -i \int e^{itz} \phi'(z) dz = i^2 t \int e^{itz} \phi(z) dz = -t\varphi(t),$$

using $\phi'(z) = -z\phi(z)$ and then integration by parts. With $\varphi(0) = 1$ the unique solution of $\varphi' = -t\varphi$ is $\varphi(t) = e^{-t^2/2}$.

In general $X = \boldsymbol{\mu} + \mathbf{A}Z$ with Z a vector of independent standard normals and $\mathbf{A}\mathbf{A}' = \boldsymbol{\Sigma}$. By Proposition P.81(iii) applied coordinatewise, $\varphi_Z(\eta) = \prod_j e^{-\eta_j^2/2} = e^{-|\eta|^2/2}$, and by (iv), $\varphi_X(\theta) = e^{i\langle \theta, \boldsymbol{\mu} \rangle} \varphi_Z(\mathbf{A}'\theta) = e^{i\langle \theta, \boldsymbol{\mu} \rangle} e^{-|\mathbf{A}'\theta|^2/2}$, with $|\mathbf{A}'\theta|^2 = \theta' \mathbf{A}\mathbf{A}' \theta = \theta' \boldsymbol{\Sigma} \theta$. $\square$

24 Lévy's Continuity Theorem

The Gaussian density on $\mathbb{R}^d$ with variance σ^2 per coordinate is $f_\sigma(x) = (2\pi\sigma^2)^{-d/2} e^{-|x|^2/(2\sigma^2)}$. For $\psi \in C_K(\mathbb{R}^d)$ put

$$\psi_\sigma(x) = \int_{\mathbb{R}^d} f_\sigma(x - y) \psi(y) dy. \quad (\text{P.17})$$

Lemma P.83 (Gaussian Fourier inversion). $f_\sigma(z) = (2\pi)^{-d} \int_{\mathbb{R}^d} e^{i\langle \theta, z \rangle} e^{-\sigma^2 |\theta|^2/2} d\theta$ for every $z \in \mathbb{R}^d$.

Proof. The integrand factors over coordinates, so it suffices to take $d = 1$. By Proposition P.82 applied to $N(0, \sigma^{-2})$, whose density is $(\sigma/\sqrt{2\pi})e^{-\sigma^2 t^2/2}$,

$$\int_{\mathbb{R}} e^{itz} \frac{\sigma}{\sqrt{2\pi}} e^{-\sigma^2 t^2/2} dt = e^{-z^2/(2\sigma^2)},$$

so $\int_{\mathbb{R}} e^{itz} e^{-\sigma^2 t^2/2} dt = \sqrt{2\pi} \sigma^{-1} e^{-z^2/(2\sigma^2)} = 2\pi f_\sigma(z)$. □

Lemma P.84 (Mollification). For $\psi \in C_K(\mathbb{R}^d)$, $\psi_\sigma \rightarrow \psi$ uniformly as $\sigma \rightarrow 0^+$.

Proof. Since $\int f_\sigma = 1$,

$$|\psi(x) - \psi_\sigma(x)| \leq \int f_\sigma(x - y) |\psi(x) - \psi(y)| dy = I_1 + I_2,$$

splitting the integral at $|y - x| \leq \delta$ and $|y - x| > \delta$. Given $\varepsilon > 0$ choose δ with $|\psi(x) - \psi(y)| \leq \varepsilon$ whenever $|x - y| \leq \delta$, possible because ψ is uniformly continuous, being continuous with compact support; then $I_1 \leq \varepsilon$. For the second, if W has density f_σ then by Markov's inequality

$$I_2 \leq 2\|\psi\|_\infty \mathbf{P}\{|W| > \delta\} \leq 2\|\psi\|_\infty \frac{\mathbf{E}|W|^2}{\delta^2} = 2\|\psi\|_\infty \frac{d\sigma^2}{\delta^2},$$

which tends to 0 as $\sigma \rightarrow 0$. Both bounds are free of x . □

Lemma P.85 (Smoothing identity). Let μ be a probability on $\mathbb{R}^d$ with characteristic function φ_μ , and let $\psi \in C_K(\mathbb{R}^d)$. Then

$$\int_{\mathbb{R}^d} \psi_\sigma d\mu = \int_{\mathbb{R}^d} \varphi_\mu(\theta) H(\theta) d\theta, \quad H(\theta) = (2\pi)^{-d} e^{-\sigma^2|\theta|^2/2} \int_{\mathbb{R}^d} \psi(y) e^{-i\langle \theta, y \rangle} dy. \quad (\text{P.18})$$

Proof. Substitute Lemma P.83 into (P.17):

$$\int \psi_\sigma d\mu = \int \int \int (2\pi)^{-d} e^{-\sigma^2|\theta|^2/2} e^{i\langle \theta, x-y \rangle} \psi(y) d\theta dy d\mu(x).$$

The modulus of the integrand is $(2\pi)^{-d}e^{-\sigma^2|\theta|^2/2}|\psi(y)|$, whose integral over θ , y and x is finite because ψ has compact support and μ is a probability. Fubini's theorem therefore permits integrating in x first, which produces $\varphi_\mu(\theta)$, and collecting the y -integral into H gives (P.18). $\square$

Lemma P.86 (Compactly supported test functions suffice). Let $\mu, \mu_1, \mu_2, \dots$ be probability measures on $\mathbb{R}^d$ with $\int \psi d\mu_n \rightarrow \int \psi d\mu$ for every $\psi \in C_K(\mathbb{R}^d)$. Then $\mu_n \rightarrow \mu$ weakly.

Proof. Fix $g \in C_b(\mathbb{R}^d)$ and $\varepsilon > 0$. Because μ is a probability there is R with $\mu\{|x| > R\} < \varepsilon$. Choose $\chi \in C_K(\mathbb{R}^d)$ with $\mathbf{1}\{|x| \leq R\} \leq \chi \leq 1$. Then $\int \chi d\mu \geq 1 - \varepsilon$, and by hypothesis $\int \chi d\mu_n \rightarrow \int \chi d\mu$, so $\limsup_n \int (1 - \chi) d\mu_n \leq \varepsilon$. Now $g\chi \in C_K(\mathbb{R}^d)$, so $\int g\chi d\mu_n \rightarrow \int g\chi d\mu$, while

$$\left| \int g(1 - \chi) d\mu_n \right| \leq \|g\|_\infty \int (1 - \chi) d\mu_n, \quad \left| \int g(1 - \chi) d\mu \right| \leq \|g\|_\infty \varepsilon.$$

Combining the three displays, $\limsup_n \left| \int g d\mu_n - \int g d\mu \right| \leq 2\|g\|_\infty \varepsilon$, and ε was arbitrary. $\square$

Theorem P.87 (Lévy's continuity theorem). Let $\mu, \mu_1, \mu_2, \dots$ be probabilities on $\mathbb{R}^d$. Then $\mu_n \rightarrow \mu$ weakly if and only if $\varphi_{\mu_n}(\theta) \rightarrow \varphi_\mu(\theta)$ for every $\theta \in \mathbb{R}^d$.

Proof. Necessity. $x \mapsto e^{i\langle \theta, x \rangle}$ has bounded continuous real and imaginary parts, so weak convergence applies to each.

Sufficiency. Assume pointwise convergence of the characteristic functions and fix $\psi \in C_K(\mathbb{R}^d)$; by Lemma P.86 it suffices to show $\int \psi d\mu_n \rightarrow \int \psi d\mu$. Let $\varepsilon > 0$. By Lemma P.84 choose σ with $\|\psi_\sigma - \psi\|_\infty \leq \varepsilon$; since every μ_n and μ is a probability,

$$\left| \int \psi d\mu_n - \int \psi d\mu \right| \leq 2\varepsilon + \left| \int \psi_\sigma d\mu_n - \int \psi_\sigma d\mu \right|.$$

For the remaining term use the smoothing identity (P.18) with the same σ for every n . The integrands $\varphi_{\mu_n} H$ converge pointwise to $\varphi_\mu H$ and are dominated in modulus by $(2\pi)^{-d} e^{-\sigma^2|\theta|^2/2} \int |\psi|$, which is integrable in θ . Dominated convergence gives $\int \psi_\sigma d\mu_n \rightarrow \int \psi_\sigma d\mu$, so the lim sup of the left side is at most 2ε . $\square$

Corollary P.88 (Uniqueness). Two probabilities on $\mathbb{R}^d$ with the same characteristic function are equal.

Proof. Apply Theorem P.87 to the constant sequence $\mu_n = \nu$: if $\varphi_\nu = \varphi_\mu$ then $\nu \rightarrow \mu$ weakly, so $\int g \, d\nu = \int g \, d\mu$ for every $g \in C_b(\mathbb{R}^d)$. Taking g to approximate the indicator of a closed set from above identifies the two measures on closed sets, hence on the Borel σ -field. $\square$

Corollary P.89 (Cramér–Wold). $X_n \xrightarrow{d} X$ in $\mathbb{R}^d$ if and only if $\langle \theta, X_n \rangle \xrightarrow{d} \langle \theta, X \rangle$ in $\mathbb{R}$ for every $\theta \in \mathbb{R}^d$.

Proof. By Theorem P.87 the left side is equivalent to $\mathbf{E}[e^{i\langle \theta, X_n \rangle}] \rightarrow \mathbf{E}[e^{i\langle \theta, X \rangle}]$ for all θ , and the right side, applied to the scalar variables $\langle \theta, X_n \rangle$ at the argument $t = 1$, says the same thing. $\square$

25 Portmanteau and the Continuous Mapping Theorem

Theorem P.90 (Portmanteau). For probabilities μ, μ_n on $\mathbb{R}^d$ the following are equivalent.

- (a) $\mu_n \rightarrow \mu$ weakly.
- (b) $\liminf_n \int f \, d\mu_n \geq \int f \, d\mu$ for every lower semi-continuous f bounded below.
- (c) $\int f \, d\mu_n \rightarrow \int f \, d\mu$ for every bounded f whose set of discontinuity points is μ -null.

Proof. (c) $\Rightarrow$ (a) is trivial, a continuous function having no discontinuities.

(a) $\Rightarrow$ (b). A lower semi-continuous f bounded below is the increasing limit of the bounded continuous functions $f_k(x) = \inf_y \{f(y) \wedge k + k|x - y|\}$. For each k ,

$$\int f_k \, d\mu = \lim_n \int f_k \, d\mu_n \leq \liminf_n \int f \, d\mu_n$$

since $f_k \leq f$; letting $k \rightarrow \infty$ and using monotone convergence on the left gives (b).

(b) $\Rightarrow$ (c). Applying (b) to $-f$ gives the upper form $\limsup_n \int f \, d\mu_n \leq \int f \, d\mu$ for upper semi-continuous f bounded above. Let f be bounded with μ -null discontinuity set, and take f_*, f^* from

Lemma P.12; they agree with f off a μ -null set, so all three have the same μ -integral. Then

$$\int f \, d\mu = \int f_* \, d\mu \leq \liminf_n \int f_* \, d\mu_n \leq \liminf_n \int f \, d\mu_n,$$

$$\int f \, d\mu = \int f^* \, d\mu \geq \limsup_n \int f^* \, d\mu_n \geq \limsup_n \int f \, d\mu_n,$$

and the two together force convergence. □

Theorem P.91 (Continuous mapping). Let $X_n \xrightarrow{d} X$ in $\mathbb{R}^d$ and let $g : \mathbb{R}^d \rightarrow \mathbb{R}^m$ be measurable with $\mathbf{P}\{X \in D_g\} = 0$, where D_g is the set of discontinuity points of g . Then $g(X_n) \xrightarrow{d} g(X)$. The same implication holds with $\xrightarrow{d}$ replaced throughout by $\xrightarrow{p}$.

Proof. Continuous g . If g is continuous everywhere the proof is one line: for $f \in C_b(\mathbb{R}^m)$ the composition $f \circ g$ lies in $C_b(\mathbb{R}^d)$, so

$$\mathbf{E}[f(g(X_n))] = \mathbf{E}[(f \circ g)(X_n)] \rightarrow \mathbf{E}[(f \circ g)(X)] = \mathbf{E}[f(g(X))].$$

General g . Let $f \in C_b(\mathbb{R}^m)$. The discontinuity set of $f \circ g$ is contained in D_g , which is μ -null for μ the law of X , and $f \circ g$ is bounded. Theorem P.90(c) applies directly and gives $\mathbf{E}[(f \circ g)(X_n)] \rightarrow \mathbf{E}[(f \circ g)(X)]$.

In probability. Suppose $X_n \xrightarrow{p} X$. By Lemma P.77 every subsequence has a further subsequence with $X_{n_k} \rightarrow X$ almost surely; on the event $\{X \notin D_g\}$, of probability one, $g(X_{n_k}) \rightarrow g(X)$. Hence $g(X_{n_k}) \xrightarrow{p} g(X)$ along that sub-subsequence, and since every subsequence admits one, the whole sequence converges in probability. $\square$

Remark. The three uses in this course are all of the everywhere-continuous kind: $\mathbf{v} \mapsto \mathbf{v}'\mathbf{A}\mathbf{v}$ for a fixed matrix, $\mathbf{A} \mapsto \mathbf{A}^{-1}$ on the open set of invertible matrices, and $x \mapsto \sqrt{x}$ on $(0, \infty)$. The measurable version is stated because the natural statement of Slutsky's theorem needs a map that is continuous only at the relevant limit.

26 Slutsky's Theorem

Theorem P.92 (Slutsky). Let $X_n \xrightarrow{d} X$ in $\mathbb{R}^d$ and $Y_n \xrightarrow{p} c$ in $\mathbb{R}^m$ with c constant. Then

- (i) $(X_n, Y_n) \xrightarrow{d} (X, c)$ in $\mathbb{R}^{d+m}$;
- (ii) $\Phi(X_n, Y_n) \xrightarrow{d} \Phi(X, c)$ for every continuous $\Phi : \mathbb{R}^d \times \mathbb{R}^m \rightarrow \mathbb{R}^l$;
- (iii) in particular, with $m = d$, $X_n + Y_n \xrightarrow{d} X + c$; with $m = 1$, $Y_n X_n \xrightarrow{d} cX$; and if $c \neq 0$, $X_n/Y_n \xrightarrow{d} X/c$.

Proof. (i) By Theorem P.87 it suffices to show the characteristic functions converge. For $\xi \in \mathbb{R}^d$ and $\eta \in \mathbb{R}^m$, split

$$\mathbf{E}[e^{i\langle \xi, X_n \rangle} e^{i\langle \eta, Y_n \rangle}] = \mathbf{E}[e^{i\langle \xi, X_n \rangle}] e^{i\langle \eta, c \rangle} + \mathbf{E}[e^{i\langle \xi, X_n \rangle} (e^{i\langle \eta, Y_n \rangle} - e^{i\langle \eta, c \rangle})].$$

The first term tends to $\varphi_X(\xi) e^{i\langle \eta, c \rangle}$, which is the characteristic function of (X, c) at (ξ, η) , because $X_n \xrightarrow{d} X$. The second is bounded in modulus by $\mathbf{E}|e^{i\langle \eta, Y_n \rangle} - e^{i\langle \eta, c \rangle}|$, and the function $y \mapsto |e^{i\langle \eta, y \rangle} - e^{i\langle \eta, c \rangle}|$ is bounded by 2 and continuous, vanishing at $y = c$; since $Y_n \xrightarrow{p} c$, Theorem P.91

makes that quantity converge in probability to 0, and being bounded it converges in mean, by Proposition P.78 and dominated convergence along almost surely convergent subsequences.

(ii) is Theorem P.91 applied to Φ and the pair from (i), and (iii) consists of three instances: $\Phi(x, y) = x + y$, $\Phi(x, y) = yx$, and $\Phi(x, y) = x/y$, the last continuous at every (x, y) with $y \neq 0$ and so covered by the measurable version of Theorem P.91, the limit (X, c) having $c \neq 0$ with probability one. $\square$

Remark. No independence between X_n and Y_n is assumed, which is what makes the theorem usable: in applications Y_n is typically an estimator computed from the same data as X_n , such as s^2 or $\hat{\tau}^2$.

27 The Lindeberg–Feller Central Limit Theorem

A **triangular array** is a doubly indexed collection $\{\xi_{nj} : 1 \leq j \leq k_n\}$ of random variables, independent within each row n . Nothing is assumed across rows, and the row entries need not be identically distributed, which is what the regression applications require, since there the summands carry non-identical weights $\mathbf{z}_i$.

Theorem P.93 (Lindeberg–Feller). Let $\{\xi_{nj}\}$ be a triangular array with $\mathbf{E}[\xi_{nj}] = 0$ and $\sum_j \text{var}(\xi_{nj}) \rightarrow 1$. If the **Lindeberg condition**

$$\sum_j \mathbf{E}[\xi_{nj}^2; |\xi_{nj}| > \varepsilon] \longrightarrow 0 \quad \text{for every } \varepsilon > 0 \quad (\text{P.19})$$

holds, then $\sum_j \xi_{nj} \xrightarrow{d} N(0, 1)$ and $\sup_j \text{var}(\xi_{nj}) \rightarrow 0$.

Proof. Write $c_{nj} = \mathbf{E}[\xi_{nj}^2]$ and $c_n = \sum_j c_{nj} \rightarrow 1$.

Step 1: the variances are uniformly negligible. For any $\varepsilon > 0$, splitting at $|\xi_{nj}| \leq \varepsilon$,

$$\sup_j c_{nj} \leq \varepsilon^2 + \sup_j \mathbf{E}[\xi_{nj}^2; |\xi_{nj}| > \varepsilon] \leq \varepsilon^2 + \sum_j \mathbf{E}[\xi_{nj}^2; |\xi_{nj}| > \varepsilon].$$

Letting $n \rightarrow \infty$ and then $\varepsilon \rightarrow 0$ gives $\sup_j c_{nj} \rightarrow 0$.

Step 2: a Gaussian comparison array. Let ζ_{nj} be independent $N(0, c_{nj})$, so that $\zeta_n = \sum_j \zeta_{nj}$ is $N(0, c_n)$ by Proposition P.81(iii) and Proposition P.82. Since $c_n \rightarrow 1$, $\varphi_{\zeta_n}(t) = e^{-c_n t^2/2} \rightarrow$

$e^{-t^2/2}$, so $\zeta_n \xrightarrow{d} N(0, 1)$ by Theorem P.87. By the same theorem it now suffices to prove, for each fixed t ,

$$\left| \prod_j \phi_{nj}(t) - \prod_j \psi_{nj}(t) \right| \longrightarrow 0, \quad (\text{P.20})$$

ϕ_{nj} and ψ_{nj} denoting the characteristic functions of ξ_{nj} and ζ_{nj} .

Step 3: reduce to individual terms. Characteristic functions have modulus at most 1, so Lemma P.17 bounds the left side of (P.20) by $\sum_j |\phi_{nj}(t) - \psi_{nj}(t)|$, and inserting $1 - \frac{1}{2}t^2 c_{nj}$,

$$\left| \prod_j \phi_{nj} - \prod_j \psi_{nj} \right| \leq \sum_j |\phi_{nj}(t) - 1 + \frac{1}{2}t^2 c_{nj}| + \sum_j |\psi_{nj}(t) - 1 + \frac{1}{2}t^2 c_{nj}|. \quad (\text{P.21})$$

Step 4: bound each term. Both ξ_{nj} and ζ_{nj} have mean zero and variance c_{nj} , so for a generic such variable ξ , taking $m = 2$ in Lemma P.16 and using $\mathbf{E}[\xi] = 0$,

$$|\mathbf{E}[e^{it\xi}] - 1 + \frac{1}{2}t^2 \mathbf{E}[\xi^2]| = |\mathbf{E}[e^{it\xi} - 1 - it\xi + \frac{1}{2}t^2 \xi^2]| \leq \mathbf{E}\left[\min\left\{t^2 \xi^2, \frac{|t|^3 |\xi|^3}{6}\right\}\right] \lesssim_t \mathbf{E}[\xi^2(1 \wedge |\xi|)],$$

the implied constant depending on t alone. Applying this to each ξ_{nj} and each ζ_{nj} turns (P.21) into

$$\left| \prod_j \phi_{nj} - \prod_j \psi_{nj} \right| \lesssim_t \sum_j \mathbf{E}[\xi_{nj}^2(1 \wedge |\xi_{nj}|)] + \sum_j \mathbf{E}[\zeta_{nj}^2(1 \wedge |\zeta_{nj}|)].$$

Step 5: both sums vanish. For the first, split at ε :

$$\sum_j \mathbf{E}[\xi_{nj}^2(1 \wedge |\xi_{nj}|)] \leq \varepsilon \sum_j c_{nj} + \sum_j \mathbf{E}[\xi_{nj}^2; |\xi_{nj}| > \varepsilon],$$

which tends to $\varepsilon \cdot 1$ by (P.19) and then to 0 as $\varepsilon \rightarrow 0$. For the second, bound $1 \wedge |\zeta| \leq |\zeta|$ and use the third absolute moment of a centred Gaussian, $\mathbf{E}|\zeta_{nj}|^3 = c_{nj}^{3/2} \mathbf{E}|Z|^3$ with Z standard normal:

$$\sum_j \mathbf{E}[\zeta_{nj}^2(1 \wedge |\zeta_{nj}|)] \leq \mathbf{E}|Z|^3 \sum_j c_{nj}^{3/2} \leq \mathbf{E}|Z|^3 c_n \sup_j c_{nj}^{1/2} \longrightarrow 0$$

by Step 1 and $c_n \rightarrow 1$. Hence (P.20) holds, and the theorem follows. $\square$

Corollary P.94 (Lindeberg–Lévy). If $X_1, X_2, \dots$ are independent and identically distributed with mean μ and finite variance $\sigma^2 > 0$, then $n^{-1/2} \sum_{i \leq n} (X_i - \mu)/\sigma \xrightarrow{d} N(0, 1)$.

Proof. Put $\xi_{nj} = (X_j - \mu)/(\sigma\sqrt{n})$ for $j \leq n$. These are independent within a row with mean 0 and $\sum_j \text{var}(\xi_{nj}) = 1$ exactly. Writing $Y = (X_1 - \mu)/\sigma$, the Lindeberg sum is

$$\sum_{j=1}^n \mathbf{E}[\xi_{nj}^2; |\xi_{nj}| > \varepsilon] = n \cdot \frac{1}{n} \mathbf{E}[Y^2; |Y| > \varepsilon\sqrt{n}] = \mathbf{E}[Y^2; |Y| > \varepsilon\sqrt{n}],$$

which tends to 0 by dominated convergence, $\mathbf{E}[Y^2] = 1$ being finite. □

Corollary P.95 (Weighted sums). Let $\varepsilon_1, \varepsilon_2, \dots$ be independent with mean 0, common variance $\tau^2 > 0$ and $\sup_i \mathbf{E}[\varepsilon_i^2; |\varepsilon_i| > a] \rightarrow 0$ as $a \rightarrow \infty$; let $\mathbf{z}_i \in \mathbb{R}^p$ be non-random with

$$n^{-1} \sum_{i \leq n} \mathbf{z}_i \mathbf{z}_i' \rightarrow \mathbf{Q} \text{ positive definite,} \quad n^{-1} \max_{i \leq n} |\mathbf{z}_i|^2 \rightarrow 0.$$

Then $n^{-1/2} \sum_{i \leq n} \mathbf{z}_i \varepsilon_i \xrightarrow{d} N(\mathbf{0}, \tau^2 \mathbf{Q})$.

Proof. By Cramér–Wold (Corollary P.89) it suffices to treat $S_n(\theta) = n^{-1/2} \sum_i \langle \theta, \mathbf{z}_i \rangle \varepsilon_i$ for fixed $\theta \neq \mathbf{0}$. Put $a_{ni} = \langle \theta, \mathbf{z}_i \rangle / \sqrt{n}$ and $\xi_{ni} = a_{ni} \varepsilon_i / v_n$ with $v_n^2 = \tau^2 n^{-1} \sum_i \langle \theta, \mathbf{z}_i \rangle^2 \rightarrow \tau^2 \theta' \mathbf{Q} \theta > 0$. Then $\mathbb{E}[\xi_{ni}] = 0$ and $\sum_i \text{var}(\xi_{ni}) = 1$. Since $|a_{ni}| \leq |\theta| n^{-1/2} \max_i |\mathbf{z}_i| =: |\theta| \delta_n$ with $\delta_n \rightarrow 0$,

$$\sum_i \mathbb{E}[\xi_{ni}^2; |\xi_{ni}| > \varepsilon] \leq \frac{1}{v_n^2} \sum_i a_{ni}^2 \mathbb{E}[\varepsilon_i^2; |\varepsilon_i| > \varepsilon v_n / (|\theta| \delta_n)] \leq \frac{\sum_i a_{ni}^2}{v_n^2} \sup_i \mathbb{E}[\varepsilon_i^2; |\varepsilon_i| > b_n]$$

with $b_n = \varepsilon v_n / (|\theta| \delta_n) \rightarrow \infty$. The prefactor tends to $1/\tau^2$ and the supremum to 0, so the Lindeberg condition holds and Theorem P.93 gives $S_n(\theta)/v_n \xrightarrow{d} N(0, 1)$, i.e. $S_n(\theta) \xrightarrow{d} N(0, \tau^2 \theta' \mathbf{Q} \theta)$, which is the law of $\langle \theta, V \rangle$ for $V \sim N(\mathbf{0}, \tau^2 \mathbf{Q})$. $\square$

Remark. The condition $n^{-1} \max_i |\mathbf{z}_i|^2 \rightarrow 0$ says no single observation dominates the design. In regression terms it is implied by $\max_i h_{ii} \rightarrow 0$, the same condition that appears in Unit 4; without

it one observation can carry a non-vanishing share of the sum and no central limit theorem holds.

Corollary P.96 (Bounded summands with unequal variances). Let $\mathbf{W}_1, \mathbf{W}_2, \dots$ be independent random vectors in $\mathbb{R}^p$ with $\mathbf{E}[\mathbf{W}_i] = \mathbf{0}$ and $|\mathbf{W}_i| \leq M$ almost surely for a constant M free of i , and suppose

$$\mathbf{F}_n = \frac{1}{n} \sum_{i \leq n} \text{var}(\mathbf{W}_i) \longrightarrow \mathbf{F} \quad \text{positive definite.}$$

Then $n^{-1/2} \sum_{i \leq n} \mathbf{W}_i \xrightarrow{d} N(\mathbf{0}, \mathbf{F})$.

Proof. By Cramér–Wold (Corollary P.89) fix $\theta \neq \mathbf{0}$ and put $\xi_{ni} = \langle \theta, \mathbf{W}_i \rangle / (\sqrt{n} v_n)$ with $v_n^2 = \theta' \mathbf{F}_n \theta \rightarrow \theta' \mathbf{F} \theta > 0$. Then $\mathbf{E}[\xi_{ni}] = 0$ and $\sum_i \text{var}(\xi_{ni}) = \theta' \mathbf{F}_n \theta / v_n^2 = 1$.

By Cauchy–Schwarz (Theorem P.34), $|\xi_{ni}| \leq |\theta| M / (\sqrt{n} v_n)$, which tends to 0 uniformly in i . So for each $\varepsilon > 0$ the event $\{|\xi_{ni}| > \varepsilon\}$ is empty for all large n and every i , and the Lindeberg sum (P.19) is exactly 0. Theorem P.93 gives $\sum_i \xi_{ni} \xrightarrow{d} N(0, 1)$, that is $n^{-1/2} \sum_i \langle \theta, \mathbf{W}_i \rangle \xrightarrow{d} N(0, \theta' \mathbf{F} \theta)$. $\square$

Remark. Corollary P.95 keeps the variance common and lets the weights vary; Corollary P.96 lets the variance vary and pays for it with a uniform bound on the summands. The score of a binary regression, $\sum_i \mathbf{x}_i(y_i - \pi_i)$, has both features and is bounded whenever the design is, so it is the second form that Unit 7 uses.

Corollary P.97 (Lyapunov). Let $\mathbf{W}_1, \mathbf{W}_2, \dots$ be independent random vectors in $\mathbb{R}^p$ with $\mathbf{E}[\mathbf{W}_i] = \mathbf{0}$ and $\sup_i \mathbf{E}|\mathbf{W}_i|^{2+\delta} \leq C < \infty$ for some $\delta > 0$, and suppose $\mathbf{F}_n = n^{-1} \sum_{i \leq n} \text{var}(\mathbf{W}_i) \rightarrow \mathbf{F}$ positive definite. Then $n^{-1/2} \sum_{i \leq n} \mathbf{W}_i \xrightarrow{d} N(\mathbf{0}, \mathbf{F})$.

Proof. As before, fix $\theta \neq \mathbf{0}$ and put $\xi_{ni} = \langle \theta, \mathbf{W}_i \rangle / (\sqrt{n} v_n)$ with $v_n^2 = \theta' \mathbf{F}_n \theta \rightarrow \theta' \mathbf{F} \theta > 0$, so that $\mathbf{E}[\xi_{ni}] = 0$ and $\sum_i \text{var}(\xi_{ni}) = 1$. On $\{|\xi_{ni}| > \varepsilon\}$ one has $1 \leq (|\xi_{ni}|/\varepsilon)^\delta$, so

$$\sum_{i \leq n} \mathbf{E}[\xi_{ni}^2; |\xi_{ni}| > \varepsilon] \leq \frac{1}{\varepsilon^\delta} \sum_{i \leq n} \mathbf{E}|\xi_{ni}|^{2+\delta} \leq \frac{|\theta|^{2+\delta} C}{\varepsilon^\delta v_n^{2+\delta}} \frac{n}{n^{1+\delta/2}},$$

using Cauchy–Schwarz (Theorem P.34) on $|\langle \theta, \mathbf{W}_i \rangle|$. The right side is $O(n^{-\delta/2}) \rightarrow 0$, so the

Lindeberg condition holds and Theorem P.93 applies. Cramér–Wold (Corollary P.89) completes the proof. $\square$

Remark. A Poisson score $\sum_i \mathbf{x}_i(y_i - \mu_i)$ is not bounded, so Unit 8 uses Corollary P.97 where Unit 7 used Corollary P.96; any $\delta > 0$ will do, and $\delta = 2$ needs only bounded fourth moments.

28 Dependent Summands

Every central limit theorem above assumes the summands independent. A regression of a series on its own past breaks that at the first step: the least squares estimator of an autoregression carries $\sum_t y_{t-1}\varepsilon_t$ in its numerator, and y_{t-1} is built from $\varepsilon_{t-1}, \varepsilon_{t-2}, \dots$, so the summands are uncorrelated but not independent and Theorem P.93 does not reach them. Three results close the gap: a law of large numbers for correlated sequences, a central limit theorem for sequences whose dependence has finite range, and a device that passes from the second to sequences whose dependence has infinite range.

Definition P.98 (Weak stationarity). A sequence $\{X_t\}_{t \in \mathbb{Z}}$ with $\mathbf{E}[X_t^2] < \infty$ is **weakly stationary** if $\mathbf{E}[X_t] = \mu$ and $\text{cov}(X_t, X_{t+k}) = \gamma(k)$ are free of t ; γ is its **autocovariance function**.

It is **strictly stationary** if the law of $(X_{t_1}, \dots, X_{t_r})$ is unchanged when every index is shifted by the same h .

Lemma P.99 (Variance of a stationary average). For a weakly stationary sequence,

$$\text{var}\left(\frac{1}{n} \sum_{t=1}^n X_t\right) = \frac{1}{n} \sum_{|k| < n} \left(1 - \frac{|k|}{n}\right) \gamma(k). \quad (\text{P.22})$$

If $\gamma(k) \rightarrow 0$ as $k \rightarrow \infty$, the right side tends to 0 and $n^{-1} \sum_{t \leq n} X_t \xrightarrow{p} \mu$.

Proof. $\text{var}(\sum_{t \leq n} X_t) = \sum_{s, t \leq n} \gamma(t - s)$, and for each k with $|k| < n$ there are exactly $n - |k|$ pairs (s, t) in $\{1, \dots, n\}^2$ with $t - s = k$; dividing by n^2 gives (P.22).

Assume $\gamma(k) \rightarrow 0$ and fix $\varepsilon > 0$. Choose K with $|\gamma(k)| \leq \varepsilon$ for $|k| > K$. Since $|\gamma(k)| \leq \gamma(0)$ by Cauchy–Schwarz (Theorem P.34) and $0 \leq 1 - |k|/n \leq 1$, the modulus of the right side of (P.22) is at most $n^{-1}(2K + 1)\gamma(0) + \varepsilon$. Letting $n \rightarrow \infty$ and then $\varepsilon \rightarrow 0$ makes it vanish, and Chebyshev’s inequality (Theorem P.57) converts that into convergence in probability. $\square$

Remark. Independence is not needed, only that the correlation between distant terms dies out. Without that the conclusion fails: a random walk has $\gamma_t(k)$ growing with t and its average does not converge at all.

Lemma P.100 (Approximation). Let S_n and $S_n^{(p)}$, $n, p \geq 1$, be real random variables such that

- (i) for each fixed p , $S_n^{(p)} \xrightarrow{d} Z_p$ as $n \rightarrow \infty$;
- (ii) $Z_p \xrightarrow{d} Z$ as $p \rightarrow \infty$;
- (iii) $\lim_{p \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbf{P}\{|S_n - S_n^{(p)}| > \varepsilon\} = 0$ for every $\varepsilon > 0$.

Then $S_n \xrightarrow{d} Z$.

Proof. By Theorem P.87 it is enough that $\mathbf{E}[e^{itS_n}] \rightarrow \mathbf{E}[e^{itZ}]$ for each fixed t . Write, for any p ,

$$|\mathbf{E}[e^{itS_n}] - \mathbf{E}[e^{itZ}]| \leq |\mathbf{E}[e^{itS_n} - e^{itS_n^{(p)}}]| + |\mathbf{E}[e^{itS_n^{(p)}}] - \mathbf{E}[e^{itZ_p}]| + |\mathbf{E}[e^{itZ_p}] - \mathbf{E}[e^{itZ}]|.$$

For the first term, $|e^{ia} - e^{ib}| = |e^{i(a-b)} - 1| \leq \min\{2, |a - b|\}$ by Lemma P.16 with $m = 0$, so for

every $\varepsilon > 0$

$$\mathbf{E}|e^{itS_n} - e^{itS_n^{(p)}}| \leq \mathbf{E}[\min\{2, |t| |S_n - S_n^{(p)}|\}] \leq |t|\varepsilon + 2 \mathbf{P}\{|S_n - S_n^{(p)}| > \varepsilon\}.$$

The second term vanishes as $n \rightarrow \infty$ by (i) and Theorem P.87. Taking $\limsup_n$ therefore leaves

$$\limsup_n |\mathbf{E}[e^{itS_n}] - \mathbf{E}[e^{itZ}]| \leq |t|\varepsilon + 2 \limsup_n \mathbf{P}\{|S_n - S_n^{(p)}| > \varepsilon\} + |\mathbf{E}[e^{itZ_p}] - \mathbf{E}[e^{itZ}]|.$$

Now let $p \rightarrow \infty$: the second term vanishes by (iii) and the third by (ii) with Theorem P.87. Finally let $\varepsilon \rightarrow 0$. □

Definition P.101 (m -dependence). $\{X_t\}$ is **m -dependent** if for every t and all finite index sets $I \subseteq \{s : s \leq t\}$ and $J \subseteq \{s : s \geq t + m + 1\}$ the random vectors $(X_s)_{s \in I}$ and $(X_s)_{s \in J}$ are independent. A 0-dependent sequence is an independent one.

Lemma P.102 (Disjoint functions of independent variables). Let $\varepsilon_1, \dots, \varepsilon_N$ be independent, let U, V be disjoint subsets of $\{1, \dots, N\}$ and let f, g be measurable functions of $(\varepsilon_u)_{u \in U}$ and

$(\varepsilon_v)_{v \in V}$ respectively. Then f and g are independent, and if both are integrable then so is fg , with $\mathbf{E}[fg] = \mathbf{E}[f] \mathbf{E}[g]$.

Proof. The law of $(\varepsilon_u)_{u \in U \cup V}$ is the product $\mu_U \otimes \mu_V$ of the laws of its two disjoint blocks. For Borel sets A, B , Fubini–Tonelli (Remark 16) gives

$$\mathbf{P}\{f \in A, g \in B\} = \iint \mathbf{1}\{f(x) \in A\} \mathbf{1}\{g(y) \in B\} d\mu_U(x) d\mu_V(y) = \mathbf{P}\{f \in A\} \mathbf{P}\{g \in B\}.$$

The same factorisation applied to $|f(x)g(y)|$ and then to $f(x)g(y)$ gives the moment statement. $\square$

Lemma P.103 (Shift functions of an independent sequence). Let $\{\varepsilon_t\}_{t \in \mathbb{Z}}$ be independent and identically distributed and put $X_t = g(\varepsilon_{t-m}, \dots, \varepsilon_t)$ for one measurable $g : \mathbb{R}^{m+1} \rightarrow \mathbb{R}$. Then $\{X_t\}$ is strictly stationary and m -dependent.

Proof. Fix indices $t_1, \dots, t_r$ and h , and let $A = \bigcup_i \{t_i - m, \dots, t_i\}$. The families $(\varepsilon_u)_{u \in A}$ and $(\varepsilon_{u+h})_{u \in A}$ have the same law, being independent with a common marginal and indexed in the same

order, and $(X_{t_1}, \dots, X_{t_r})$ and $(X_{t_1+h}, \dots, X_{t_r+h})$ are the image of the two families under the same map. That is strict stationarity. For m -dependence, $(X_s)_{s \in I}$ with $I \subseteq \{s \leq t\}$ finite is a function of $(\varepsilon_u)_{u \in U}$ with $U \subseteq \{u \leq t\}$ finite, while $(X_s)_{s \in J}$ with $J \subseteq \{s \geq t + m + 1\}$ finite is a function of $(\varepsilon_u)_{u \in V}$ with $V \subseteq \{u \geq t + 1\}$; the two index sets are disjoint, so Lemma P.102 applies. $\square$

Theorem P.104 (Central limit theorem for m -dependent sequences). Let $\{X_t\}$ be strictly stationary and m -dependent with $\mathbf{E}[X_t] = 0$ and $\mathbf{E}[X_t^2] < \infty$, and write $\gamma(k) = \mathbf{E}[X_t X_{t+k}]$. Then $\gamma(k) = 0$ for $|k| > m$,

$$\sigma^2 = \gamma(0) + 2 \sum_{k=1}^m \gamma(k) \geq 0, \quad (\text{P.23})$$

and $n^{-1/2} \sum_{t=1}^n X_t \xrightarrow{d} N(0, \sigma^2)$, where $N(0, 0)$ is read as the point mass at 0.

Proof. That $\gamma(k) = 0$ for $|k| > m$ is Definition P.101 with $I = \{t\}$ and $J = \{t + k\}$. Hence, by

Lemma P.99 applied to a block of l consecutive terms,

$$\text{var}\left(\sum_{t=1}^l X_t\right) = \sum_{|k|<l} (l - |k|)\gamma(k) = \sum_{|k|\leq m\wedge(l-1)} (l - |k|)\gamma(k). \quad (\text{P.24})$$

Taking $l \rightarrow \infty$ after dividing by l shows σ^2 is a limit of variances, so $\sigma^2 \geq 0$. Put $C = \sum_{|k|\leq m} |\gamma(k)|$, so $\text{var}(\sum_{t\leq l} X_t) \leq lC$ for every l .

Blocks. Fix an integer $p > m$ and let $r = \lfloor n/(p+m) \rfloor$. Cut $\{1, \dots, n\}$ into r consecutive stretches of length $p+m$ followed by a remainder, and inside the i th stretch let U_i be the sum of the first p terms and V_i the sum of the last m ; let R_n be the sum over the remainder, which has fewer than $p+m$ terms. Then

$$\sum_{t=1}^n X_t = \sum_{i=1}^r U_i + \sum_{i=1}^r V_i + R_n.$$

The big blocks are independent and identically distributed. The index set of U_i ends at $(i-1)(p+m) + p$ and that of U_{i+1} begins $m+1$ places later, so Definition P.101 makes

$(U_1, \dots, U_i)$ independent of $(U_{i+1}, \dots, U_r)$ for each i , and iterating gives full independence; strict stationarity makes them identically distributed. The same argument, with the gap now the big block of length $p > m$, applies to $V_1, \dots, V_r$.

Limit of the big-block sum. By (P.24) with $l = p > m$,

$$\text{var}(U_1) = \sum_{|k| \leq m} (p - |k|) \gamma(k) = p \sigma^2 - c, \quad c = \sum_{|k| \leq m} |k| \gamma(k),$$

which is free of n . If $\text{var}(U_1) = 0$ then each U_i vanishes almost surely and $S_n^{(p)} := n^{-1/2} \sum_i U_i$ is identically 0; otherwise Corollary P.94 gives $r^{-1/2} \sum_i U_i \xrightarrow{d} N(0, \text{var}(U_1))$. Since $r/n \rightarrow (p+m)^{-1}$, Theorem P.92 yields in both cases

$$S_n^{(p)} = \sqrt{r/n} \left(r^{-1/2} \sum_{i \leq r} U_i \right) \xrightarrow{d} Z_p \sim N(0, \sigma_p^2), \quad \sigma_p^2 = \frac{p \sigma^2 - c}{p + m}.$$

As $p \rightarrow \infty$, $\sigma_p^2 \rightarrow \sigma^2$, so $\mathbf{E}[e^{itZ_p}] = e^{-\sigma_p^2 t^2/2} \rightarrow e^{-\sigma^2 t^2/2}$ and $Z_p \xrightarrow{d} N(0, \sigma^2)$ by Theorem P.87.

The small blocks and the remainder are negligible. The V_i are independent with mean 0 and, by (P.24) with $l = m$, $\text{var}(V_1) \leq mC$; and $\text{var}(R_n) \leq (p + m)C$. Hence

$$\mathbf{E}[(S_n - S_n^{(p)})^2] = \frac{1}{n} \mathbf{E}\left[\left(\sum_i V_i + R_n\right)^2\right] \leq \frac{2}{n} \{r \text{ var}(V_1) + \text{var}(R_n)\} \leq \frac{2mC}{p + m} + \frac{2(p + m)C}{n},$$

using $(a + b)^2 \leq 2a^2 + 2b^2$ and $r \leq n/(p + m)$. By Chebyshev's inequality (Theorem P.57), $\limsup_n \mathbf{P}\{|S_n - S_n^{(p)}| > \varepsilon\} \leq 2mC/\{(p + m)\varepsilon^2\}$, which tends to 0 as $p \rightarrow \infty$. Lemma P.100 completes the proof. $\square$

The last step of the section removes the restriction to finite-range dependence in the one case the course needs, a product of a present innovation with a linear function of the past. Everything is phrased inside L^2 , so no conditioning on infinite families is required.

Definition P.105 (The past space). Let $\{\varepsilon_t\}_{t \in \mathbb{Z}}$ be independent and identically distributed with mean 0 and variance $\tau^2 \in (0, \infty)$. Let $\mathcal{L}_t$ be the closure in L^2 (Definition P.62) of the set of square-integrable measurable functions of finitely many ε_u with $u \leq t$.

Each $\mathcal{L}_t$ is a closed subspace of L^2 and $\mathcal{L}_s \subseteq \mathcal{L}_t$ for $s \leq t$. If $(c_j)_{j \geq 0}$ is square summable then the

partial sums of $\sum_{j \geq 0} c_j \varepsilon_{t-j}$ are Cauchy in L^2 . Indeed, the increments are orthogonal and

$$\mathbb{E} \left[\left(\sum_{M < j \leq N} c_j \varepsilon_{t-j} \right)^2 \right] = \tau^2 \sum_{M < j \leq N} c_j^2.$$

Thus the series has an L^2 limit and that limit lies in $\mathcal{L}_t$.

Lemma P.106 (Innovation orthogonality). Let $v \in \mathcal{L}_{t-1}$ and put $X = v \varepsilon_t$. Then $X \in \mathcal{L}_t$ and

$$\mathbb{E}[X] = 0, \quad \mathbb{E}[X^2] = \tau^2 \mathbb{E}[v^2], \quad \mathbb{E}[ZX] = 0 \text{ for every } Z \in \mathcal{L}_{t-1}.$$

Proof. Choose $v^{(M)} \rightarrow v$ in L^2 , each a square-integrable measurable function of finitely many ε_u with $u \leq t-1$. For fixed M , Lemma P.102 makes $v^{(M)}$ and ε_t independent, so $v^{(M)} \varepsilon_t$ is square integrable with

$$\mathbb{E}[v^{(M)} \varepsilon_t] = 0, \quad \mathbb{E}[(v^{(M)} \varepsilon_t)^2] = \tau^2 \mathbb{E}[(v^{(M)})^2].$$

Applying the second identity to $v^{(M)} - v^{(M')}$ shows $\{v^{(M)} \varepsilon_t\}_M$ is Cauchy in L^2 , hence convergent there; its limit is $v \varepsilon_t$, because L^2 convergence implies convergence in probability by Chebyshev's inequality (Theorem P.57) and therefore almost sure convergence along a subsequence (Lemma P.77),

along which $v^{(M)}\varepsilon_t \rightarrow v\varepsilon_t$. So $X \in \mathcal{L}_t$ and the two displayed identities pass to the limit. For the third claim take $Z^{(M')} \rightarrow Z$ in L^2 of the same kind. Then $Z^{(M')}v^{(M)}$ is a measurable function of finitely many ε_u with $u \leq t-1$, and it is integrable by Cauchy–Schwarz (Theorem P.34), so Lemma P.102 gives $\mathbf{E}[Z^{(M')}v^{(M)}\varepsilon_t] = \mathbf{E}[Z^{(M')}v^{(M)}] \mathbf{E}[\varepsilon_t] = 0$. Letting $M, M' \rightarrow \infty$ and using Cauchy–Schwarz once more on $\mathbf{E}[Z^{(M')}v^{(M)}\varepsilon_t - ZX]$ gives $\mathbf{E}[ZX] = 0$. $\square$

Corollary P.107 (Orthogonality across time). If $v_{t-1} \in \mathcal{L}_{t-1}$ for every t and $X_t = v_{t-1}\varepsilon_t$, then $\mathbf{E}[X_t] = 0$, $\mathbf{E}[X_sX_t] = 0$ for $s \neq t$, and

$$\mathbf{E}\left[\left(\sum_{t=1}^n X_t\right)^2\right] = \tau^2 \sum_{t=1}^n \mathbf{E}[v_{t-1}^2]. \quad (\text{P.25})$$

Proof. For $s < t$, $X_s \in \mathcal{L}_s \subseteq \mathcal{L}_{t-1}$ by Lemma P.106, so the third statement of that lemma gives $\mathbf{E}[X_sX_t] = 0$. Expanding the square and using $\mathbf{E}[X_t^2] = \tau^2 \mathbf{E}[v_{t-1}^2]$ gives (P.25). $\square$

Theorem P.108 (Central limit theorem for autoregressive scores). Let $\{\varepsilon_t\}_{t \in \mathbb{Z}}$ be independent and identically distributed with mean 0 and variance $\tau^2 \in (0, \infty)$, let $\mathbf{a}_0, \mathbf{a}_1, \dots$ lie in $\mathbb{R}^d$ with $\sum_{j \geq 0} |\mathbf{a}_j|^2 < \infty$, and let

$$\mathbf{u}_t = \sum_{j \geq 0} \mathbf{a}_j \varepsilon_{t-j}$$

be the L^2 limit of its partial sums. Then $\mathbf{\Gamma} := \mathbf{E}[\mathbf{u}_t \mathbf{u}_t'] = \tau^2 \sum_{j \geq 0} \mathbf{a}_j \mathbf{a}_j'$ is free of t and

$$\frac{1}{\sqrt{n}} \sum_{t=1}^n \mathbf{u}_{t-1} \varepsilon_t \xrightarrow{d} N(\mathbf{0}, \tau^2 \mathbf{\Gamma}). \quad (\text{P.26})$$

Proof. The limit variance. Writing $\mathbf{u}_t^{(m)} = \sum_{j < m} \mathbf{a}_j \varepsilon_{t-j}$, the ε being uncorrelated with mean 0 gives $\mathbf{E}[\mathbf{u}_t^{(m)} \mathbf{u}_t^{(m)'}] = \tau^2 \sum_{j < m} \mathbf{a}_j \mathbf{a}_j'$, and $\mathbf{u}_t^{(m)} \rightarrow \mathbf{u}_t$ in L^2 , so Cauchy–Schwarz (Theorem P.34) passes the identity to the limit.

Reduction to one dimension. By Cramér–Wold (Corollary P.89) it suffices to show, for each fixed $\theta \in \mathbb{R}^d$, that $n^{-1/2} \sum_t \langle \theta, \mathbf{u}_{t-1} \rangle \varepsilon_t \xrightarrow{d} N(0, \tau^2 \theta' \mathbf{\Gamma} \theta)$. Put $a_j = \langle \theta, \mathbf{a}_j \rangle$ and $u_t = \langle \theta, \mathbf{u}_t \rangle =$

$\sum_j a_j \varepsilon_{t-j}$, a square-summable combination with $\mathbf{E}[u_t^2] = \tau^2 \sum_j a_j^2 = \theta' \mathbf{\Gamma} \theta$. So it is enough to prove the scalar statement

$$S_n = \frac{1}{\sqrt{n}} \sum_{t=1}^n u_{t-1} \varepsilon_t \xrightarrow{d} N\left(0, \tau^4 \sum_{j \geq 0} a_j^2\right).$$

Truncation. Put $u_{t-1}^{(m)} = \sum_{j < m} a_j \varepsilon_{t-1-j}$, $X_t^{(m)} = u_{t-1}^{(m)} \varepsilon_t$ and $S_n^{(m)} = n^{-1/2} \sum_{t \leq n} X_t^{(m)}$. Since $X_t^{(m)} = g(\varepsilon_{t-m}, \dots, \varepsilon_t)$ with $g(x_0, \dots, x_m) = x_m \sum_{j < m} a_j x_{m-1-j}$, Lemma P.103 makes $\{X_t^{(m)}\}$ strictly stationary and m -dependent. Corollary P.107 gives $\mathbf{E}[X_t^{(m)}] = 0$, $\gamma^{(m)}(k) = 0$ for $k \neq 0$ and $\gamma^{(m)}(0) = \tau^2 \mathbf{E}[(u^{(m)})^2] = \tau^4 \sum_{j < m} a_j^2$. So Theorem P.104 applies with $\sigma^2 = \gamma^{(m)}(0)$:

$$S_n^{(m)} \xrightarrow{d} Z_m \sim N\left(0, \tau^4 \sum_{j < m} a_j^2\right) \quad (n \rightarrow \infty), \quad Z_m \xrightarrow{d} N\left(0, \tau^4 \sum_{j \geq 0} a_j^2\right) \quad (m \rightarrow \infty),$$

the second by Theorem P.87 as in the previous proof.

Negligibility. The difference is $X_t - X_t^{(m)} = w_{t-1}^{(m)} \varepsilon_t$ with $w_{t-1}^{(m)} = \sum_{j \geq m} a_j \varepsilon_{t-1-j} \in \mathcal{L}_{t-1}$ and

$\mathbf{E}[(w_{t-1}^{(m)})^2] = \tau^2 \sum_{j \geq m} a_j^2$, free of t . By Corollary P.107,

$$\mathbf{E}[(S_n - S_n^{(m)})^2] = \frac{1}{n} \cdot n \tau^2 \mathbf{E}[(w^{(m)})^2] = \tau^4 \sum_{j \geq m} a_j^2,$$

free of n and tending to 0 as $m \rightarrow \infty$. Chebyshev's inequality (Theorem P.57) turns this into hypothesis (iii) of Lemma P.100, which now gives the scalar statement. $\square$

Remark. The summands $u_{t-1}\varepsilon_t$ are uncorrelated by Corollary P.107 but far from independent, and $\text{var}(S_n) = \tau^2 \mathbf{E}[u^2]$ exactly, for every n . What the theorem adds is that the limit law is the normal one with that variance. Two of the course's statistics are of this form: with $a_j = \phi^j$ it is the score of an $AR(1)$ least squares fit, and with $a_j = \mathbf{1}\{j = k - 1\}$ it is the numerator of the lag k sample autocorrelation of white noise.

29 The Delta Method

Theorem P.109 (Delta method). Let $r_n \rightarrow \infty$ and $r_n(T_n - \vartheta) \xrightarrow{d} T$ in $\mathbb{R}^d$, and let $g : \mathbb{R}^d \rightarrow \mathbb{R}^m$ be differentiable at ϑ with Jacobian $\mathbf{D}_g(\vartheta)$. Then

$$r_n(g(T_n) - g(\vartheta)) \xrightarrow{d} \mathbf{D}_g(\vartheta) T.$$

In particular, if $\sqrt{n}(T_n - \vartheta) \xrightarrow{d} N(\mathbf{0}, \Sigma)$ then $\sqrt{n}\{g(T_n) - g(\vartheta)\} \xrightarrow{d} N(\mathbf{0}, \mathbf{D}_g \Sigma \mathbf{D}_g')$.

Proof. Differentiability at ϑ means $g(\vartheta + h) - g(\vartheta) = \mathbf{D}_g(\vartheta)h + R(h)$ with $|R(h)|/|h| \rightarrow 0$ as $h \rightarrow 0$. Since $r_n(T_n - \vartheta)$ converges in distribution and $r_n \rightarrow \infty$, Proposition P.79 and Theorem P.92(iii) give $T_n - \vartheta = r_n^{-1} \cdot r_n(T_n - \vartheta) \xrightarrow{p} \mathbf{0}$. Writing $H_n = T_n - \vartheta$,

$$r_n(g(T_n) - g(\vartheta)) = \mathbf{D}_g(\vartheta) r_n H_n + r_n R(H_n).$$

The first term converges in distribution to $\mathbf{D}_g(\vartheta)T$ by Theorem P.91. For the second, define $\rho(h) = |R(h)|/|h|$ for $h \neq \mathbf{0}$ and $\rho(\mathbf{0}) = 0$, so ρ is continuous at $\mathbf{0}$ and $|r_n R(H_n)| = \rho(H_n) |r_n H_n|$.

Here $\rho(H_n) \xrightarrow{p} 0$ by Theorem P.91 and $|r_n H_n| \xrightarrow{d} |T|$, so the product converges in probability to 0 by Theorem P.92(iii). One more application of Slutsky adds the two terms. $\square$

Part IV

Results Used in the Course

30 Maximum Likelihood

Units 7 and 8 fit their models by maximum likelihood and test with the Wald and likelihood ratio statistics. The results are collected here for independent but not identically distributed observations, which is what a regression design produces, and under a concavity hypothesis, which every model in those units satisfies.

30.1 Score and information

Definition P.110 (Log-likelihood, score, information). Let $\{f(\cdot; \boldsymbol{\theta}) : \boldsymbol{\theta} \in \Theta\}$ be a family of densities with respect to a measure μ on $\mathcal{X}$, with $\Theta \subseteq \mathbb{R}^p$ open. For an observation Y the **log-likelihood** is $\ell(\boldsymbol{\theta}) = \ln f(Y; \boldsymbol{\theta})$, the **score** is $\mathbf{U}(\boldsymbol{\theta}) = \nabla \ell(\boldsymbol{\theta})$, and the **information** at $\boldsymbol{\theta}$ is $\mathcal{I}(\boldsymbol{\theta}) = \text{var}_{\boldsymbol{\theta}}(\mathbf{U}(\boldsymbol{\theta}))$.

Lemma P.111 (Score identities). Suppose $f(y; \boldsymbol{\theta}) > 0$ for all $y, \boldsymbol{\theta}$, that $\boldsymbol{\theta} \mapsto f(y; \boldsymbol{\theta})$ is twice continuously differentiable on Θ , and that there are μ -integrable g_1, g_2 with

$$\left| \frac{\partial f}{\partial \theta_j}(y; \boldsymbol{\theta}) \right| \leq g_1(y), \quad \left| \frac{\partial^2 f}{\partial \theta_j \partial \theta_k}(y; \boldsymbol{\theta}) \right| \leq g_2(y)$$

for every j, k , every y and every $\boldsymbol{\theta} \in \Theta$. Then, with Y distributed according to $f(\cdot; \boldsymbol{\theta})$,

$$\mathbb{E}_{\boldsymbol{\theta}}[\mathbf{U}(\boldsymbol{\theta})] = \mathbf{0}, \quad \mathcal{I}(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{\theta}}[\mathbf{U}(\boldsymbol{\theta})\mathbf{U}(\boldsymbol{\theta})'] = -\mathbb{E}_{\boldsymbol{\theta}}[\nabla^2 \ell(\boldsymbol{\theta})]. \quad (\text{P.27})$$

Proof. Apply Theorem P.8 to $\int f(y; \boldsymbol{\theta}) \, d\mu(y) = 1$, one coordinate of $\boldsymbol{\theta}$ at a time with the others fixed, the hypothesis on g_1 supplying the domination:

$$\mathbf{0} = \nabla \int f(y; \boldsymbol{\theta}) \, d\mu(y) = \int \nabla f(y; \boldsymbol{\theta}) \, d\mu(y) = \int \frac{\nabla f(y; \boldsymbol{\theta})}{f(y; \boldsymbol{\theta})} f(y; \boldsymbol{\theta}) \, d\mu(y) = \mathbf{E}_{\boldsymbol{\theta}}[\mathbf{U}(\boldsymbol{\theta})],$$

since $\mathbf{U} = \nabla \ln f = \nabla f / f$. Differentiating once more, with g_2 supplying the domination,

$$\mathbf{0} = \int \nabla^2 f(y; \boldsymbol{\theta}) \, d\mu(y).$$

The quotient rule gives $\nabla^2 \ln f = \nabla^2 f / f - (\nabla f / f)(\nabla f / f)' = \nabla^2 f / f - \mathbf{U}\mathbf{U}'$, so taking expectations,

$$\mathbf{E}_{\boldsymbol{\theta}}[\nabla^2 \ell(\boldsymbol{\theta})] = \int \nabla^2 f(y; \boldsymbol{\theta}) \, d\mu(y) - \mathbf{E}_{\boldsymbol{\theta}}[\mathbf{U}\mathbf{U}'] = -\mathbf{E}_{\boldsymbol{\theta}}[\mathbf{U}\mathbf{U}'].$$

Finally $\text{var}(\mathbf{U}) = \mathbf{E}[\mathbf{U}\mathbf{U}']$ because $\mathbf{E}[\mathbf{U}] = \mathbf{0}$. □

Remark. When μ is counting measure on a finite set, as for the Bernoulli and multinomial families of Unit 7, the integrals are finite sums and the domination hypothesis is automatic. The equality of the two expressions for $\mathcal{I}$ in (P.27) is the **information equality**; it is a property of the model at the true parameter, not an identity in $\boldsymbol{\theta}$ for a misspecified one.

30.2 The maximum likelihood estimator

Theorem P.112 (Concave maximum likelihood). Let $Y_1, Y_2, \dots$ be independent, let $\Theta \subseteq \mathbb{R}^p$ be open and convex, let $\boldsymbol{\theta}_0 \in \Theta$, and let

$$\ell_n(\boldsymbol{\theta}) = \sum_{i=1}^n \ell_i(\boldsymbol{\theta}), \quad \mathbf{U}_n = \nabla \ell_n(\boldsymbol{\theta}_0), \quad \mathbf{H}_n(\boldsymbol{\theta}) = -\nabla^2 \ell_n(\boldsymbol{\theta}).$$

Assume:

- (L1) each ℓ_i is concave and twice continuously differentiable on Θ , whatever the value of Y_i , and $B(\boldsymbol{\theta}_0, \delta_0) \subseteq \Theta$ for some $\delta_0 > 0$;
- (L2) $\mathbb{E}[\nabla \ell_i(\boldsymbol{\theta}_0)] = \mathbf{0}$, $\mathbf{F}_n = n^{-1} \sum_{i \leq n} \text{var}(\nabla \ell_i(\boldsymbol{\theta}_0)) \rightarrow \mathbf{F}$ positive definite, and $n^{-1/2} \mathbf{U}_n \xrightarrow{d} N(\mathbf{0}, \mathbf{F})$;

(L3) $n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0) \xrightarrow{p} \mathbf{G}$ positive definite, and there is a constant L with $\|\mathbf{H}_n(\boldsymbol{\theta}) - \mathbf{H}_n(\boldsymbol{\theta}_0)\| \leq nL|\boldsymbol{\theta} - \boldsymbol{\theta}_0|$ almost surely for every $|\boldsymbol{\theta} - \boldsymbol{\theta}_0| \leq \delta_0$.

Then, writing $\lambda = \lambda_{\min}(\mathbf{G}) > 0$:

- (i) with probability tending to one ℓ_n has a maximiser $\hat{\boldsymbol{\theta}}_n$ over Θ , and it is unique whenever some ℓ_i is strictly concave;
- (ii) $\hat{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$;
- (iii) $\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{G}^{-1}\mathbf{F}\mathbf{G}^{-1})$;
- (iv) if moreover $\mathbf{F} = \mathbf{G}$, then $\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{F}^{-1})$ and $2\{\ell_n(\hat{\boldsymbol{\theta}}_n) - \ell_n(\boldsymbol{\theta}_0)\} \xrightarrow{d} \chi_p^2$;
- (v) $n^{-1}\mathbf{H}_n(\hat{\boldsymbol{\theta}}_n) \xrightarrow{p} \mathbf{G}$.

Remark. $\mathbf{G}$ is the limiting curvature of the criterion and $\mathbf{F}$ the limiting variance of its gradient. When ℓ_n is the log-likelihood of a correctly specified model the two agree, by the information equality of Lemma P.111, and (iv) applies. When only the mean is correctly specified they need not, and (iii) leaves the **sandwich** $\mathbf{G}^{-1}\mathbf{F}\mathbf{G}^{-1}$, which is what Unit 8 needs for overdispersed counts.

Proof. Step 1: a curvature floor near $\boldsymbol{\theta}_0$. Put $\delta_1 = \min\{\delta_0, \lambda/(4L)\}$ and $A_n = \{\lambda_{\min}(n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0)) > \frac{3}{4}\lambda\}$. The smallest eigenvalue of a symmetric matrix is $\min_{|\mathbf{u}|=1} \mathbf{u}'\mathbf{A}\mathbf{u}$ by Corollary P.38, hence 1-Lipschitz in the operator norm, so $\mathbf{P}(A_n) \rightarrow 1$ by (L3). On A_n , for $|\boldsymbol{\theta} - \boldsymbol{\theta}_0| \leq \delta_1$ and $|\mathbf{u}| = 1$,

$$\mathbf{u}'\frac{1}{n}\mathbf{H}_n(\boldsymbol{\theta})\mathbf{u} \geq \mathbf{u}'\frac{1}{n}\mathbf{H}_n(\boldsymbol{\theta}_0)\mathbf{u} - \left\|\frac{1}{n}\mathbf{H}_n(\boldsymbol{\theta}) - \frac{1}{n}\mathbf{H}_n(\boldsymbol{\theta}_0)\right\| \geq \frac{3}{4}\lambda - L\delta_1 \geq \frac{1}{2}\lambda. \quad (\text{P.28})$$

Step 2: the score is of order $\sqrt{n}$. The summands of $\mathbf{U}_n$ are independent with mean zero by (L2), so $\mathbf{E}|\mathbf{U}_n|^2 = \sum_i \text{tr} \text{var}(\nabla \ell_i(\boldsymbol{\theta}_0)) = n \text{tr} \mathbf{F}_n$, and Chebyshev's inequality (Theorem P.57) gives

$$\mathbf{P}\{|\mathbf{U}_n| > K\sqrt{n}\} \leq \text{tr} \mathbf{F}_n / K^2. \quad (\text{P.29})$$

Step 3: consistency. Fix $\eta > 0$ and $0 < \delta \leq \delta_1$, and choose K with $\sup_n \text{tr} \mathbf{F}_n / K^2 < \eta$, possible since $\text{tr} \mathbf{F}_n \rightarrow \text{tr} \mathbf{F}$. Let $C_n = A_n \cap \{|\mathbf{U}_n| \leq K\sqrt{n}\}$, so $\liminf \mathbf{P}(C_n) \geq 1 - \eta$. For a unit vector $\mathbf{u}$ apply Taylor's theorem (Theorem P.15, $m = 1$) to $g(t) = \ell_n(\boldsymbol{\theta}_0 + t\delta\mathbf{u})$: for some $s \in (0, 1)$,

$$\ell_n(\boldsymbol{\theta}_0 + \delta\mathbf{u}) - \ell_n(\boldsymbol{\theta}_0) = \delta \mathbf{u}'\mathbf{U}_n - \frac{1}{2}\delta^2 \mathbf{u}'\mathbf{H}_n(\boldsymbol{\theta}_0 + s\delta\mathbf{u})\mathbf{u} \leq \delta K\sqrt{n} - \frac{1}{4}n\lambda\delta^2$$

on C_n , using Cauchy–Schwarz and (P.28). The right side is negative once $n > (4K/(\lambda\delta))^2$, and the bound does not depend on $\mathbf{u}$.

Concavity extends this beyond the sphere: if $\boldsymbol{\theta} \in \Theta$ has $|\boldsymbol{\theta} - \boldsymbol{\theta}_0| = r > \delta$ then $\boldsymbol{\theta}_\delta = (1 - \delta/r)\boldsymbol{\theta}_0 + (\delta/r)\boldsymbol{\theta}$ lies on the sphere, and concavity gives $\ell_n(\boldsymbol{\theta}_\delta) \geq (1 - \delta/r)\ell_n(\boldsymbol{\theta}_0) + (\delta/r)\ell_n(\boldsymbol{\theta})$; since $\ell_n(\boldsymbol{\theta}_\delta) < \ell_n(\boldsymbol{\theta}_0)$ this forces $\ell_n(\boldsymbol{\theta}) < \ell_n(\boldsymbol{\theta}_0)$. So on C_n and for n large, ℓ_n is strictly below $\ell_n(\boldsymbol{\theta}_0)$ outside the open ball of radius δ .

The closed ball is compact and ℓ_n continuous, so a maximum over it is attained (Theorem P.14), and by the previous paragraph it is a maximum over all of Θ and lies in the open ball. This proves (i), and $\liminf \mathbf{P}\{|\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0| < \delta\} \geq 1 - \eta$ with η and δ arbitrary proves (ii). If some ℓ_i is strictly concave then so is ℓ_n , and a strictly concave function has at most one maximiser.

Step 4: the linearisation. On C_n and for large n the maximiser is interior, so $\nabla \ell_n(\hat{\boldsymbol{\theta}}_n) = \mathbf{0}$. Write $\mathbf{h} = \hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0$. Applying the mean value theorem (Corollary P.7) to $t \mapsto \partial_j \ell_n(\boldsymbol{\theta}_0 + t\mathbf{h})$ gives, for each j , an $s_j \in (0, 1)$ with

$$0 - (\mathbf{U}_n)_j = (\nabla \ell_n(\hat{\boldsymbol{\theta}}_n) - \nabla \ell_n(\boldsymbol{\theta}_0))_j = -(\mathbf{H}_n(\boldsymbol{\theta}_0 + s_j \mathbf{h}) \mathbf{h})_j,$$

that is $\mathbf{U}_n = \tilde{\mathbf{H}}_n \mathbf{h}$, where row j of $\tilde{\mathbf{H}}_n$ is row j of $\mathbf{H}_n(\boldsymbol{\theta}_0 + s_j \mathbf{h})$. Each row of $\tilde{\mathbf{H}}_n - \mathbf{H}_n(\boldsymbol{\theta}_0)$ is a row

of a matrix of operator norm at most $nL|\mathbf{h}|$ by (L3), so $\|n^{-1}\tilde{\mathbf{H}}_n - n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0)\| \leq \sqrt{p}L|\mathbf{h}|$, which tends to 0 in probability by (ii). With (L3) again,

$$\mathbf{G}_n := n^{-1}\tilde{\mathbf{H}}_n \xrightarrow{p} \mathbf{G}. \quad (\text{P.30})$$

The same argument with $\mathbf{H}_n(\hat{\boldsymbol{\theta}}_n)$ in place of $\tilde{\mathbf{H}}_n$ gives (v).

Step 5: asymptotic normality. Write $\mathbf{Z}_n = n^{-1/2}\mathbf{U}_n$, which converges in distribution to $N(\mathbf{0}, \mathbf{F})$ by (L2). On C_n , $\mathbf{G}_n$ is invertible by (P.28) and $\sqrt{n}\mathbf{h} = \mathbf{G}_n^{-1}\mathbf{Z}_n$. Matrix inversion is continuous at $\mathbf{G}$, so $\mathbf{G}_n^{-1} \xrightarrow{p} \mathbf{G}^{-1}$ by Theorem P.91, and Slutsky (Theorem P.92) with the continuous mapping theorem gives $\sqrt{n}\mathbf{h} \xrightarrow{d} \mathbf{G}^{-1}Z$ with $Z \sim N(\mathbf{0}, \mathbf{F})$. By Theorem P.68 that law is $N(\mathbf{0}, \mathbf{G}^{-1}\mathbf{F}\mathbf{G}^{-1})$, which is (iii); since $\mathbf{P}(C_n)$ can be made arbitrarily close to 1, the conclusion holds unconditionally. If $\mathbf{F} = \mathbf{G}$ the variance collapses to $\mathbf{F}^{-1}$.

Step 6: the likelihood ratio at $\boldsymbol{\theta}_0$. Taylor's theorem applied to $t \mapsto \ell_n(\boldsymbol{\theta}_0 + t\mathbf{h})$ gives an $s \in (0, 1)$ with

$$2\{\ell_n(\hat{\boldsymbol{\theta}}_n) - \ell_n(\boldsymbol{\theta}_0)\} = 2\mathbf{U}_n'\mathbf{h} - \mathbf{h}'\mathbf{H}_n(\boldsymbol{\theta}_0 + s\mathbf{h})\mathbf{h} = 2\mathbf{Z}_n'\mathbf{G}_n^{-1}\mathbf{Z}_n - (\mathbf{G}_n^{-1}\mathbf{Z}_n)'\mathbf{K}_n(\mathbf{G}_n^{-1}\mathbf{Z}_n),$$

where $\mathbf{K}_n = n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0 + s\mathbf{h}) \xrightarrow{p} \mathbf{G}$ by the argument of Step 4. Slutsky and the continuous mapping

theorem give the limit $2Z'\mathbf{G}^{-1}Z - Z'\mathbf{G}^{-1}\mathbf{G}\mathbf{G}^{-1}Z = Z'\mathbf{G}^{-1}Z$. Under $\mathbf{F} = \mathbf{G}$ this is $Z'\mathbf{F}^{-1}Z$ with $Z \sim N(\mathbf{0}, \mathbf{F})$, which is χ_p^2 by Theorem P.115, completing (iv). $\square$

Corollary P.113 (Wald). Under Theorem P.112 with $\mathbf{F} = \mathbf{G}$, let $\mathbf{R}$ be a fixed $q \times p$ matrix of rank q and $\hat{\mathbf{G}}_n = n^{-1}\mathbf{H}_n(\hat{\boldsymbol{\theta}}_n)$. If $\mathbf{R}\boldsymbol{\theta}_0 = \mathbf{r}$ then

$$W_n = (\mathbf{R}\hat{\boldsymbol{\theta}}_n - \mathbf{r})' \left\{ \frac{1}{n} \mathbf{R} \hat{\mathbf{G}}_n^{-1} \mathbf{R}' \right\}^{-1} (\mathbf{R}\hat{\boldsymbol{\theta}}_n - \mathbf{r}) \xrightarrow{d} \chi_q^2.$$

The same holds without $\mathbf{F} = \mathbf{G}$ if $\hat{\mathbf{G}}_n^{-1}$ is replaced by $\hat{\mathbf{G}}_n^{-1} \hat{\mathbf{F}}_n \hat{\mathbf{G}}_n^{-1}$ for any $\hat{\mathbf{F}}_n \xrightarrow{p} \mathbf{F}$.

Proof. By (iv) and Theorem P.68, $\sqrt{n}(\mathbf{R}\hat{\boldsymbol{\theta}}_n - \mathbf{r}) = \mathbf{R}\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{R}\mathbf{F}^{-1}\mathbf{R}')$, and $\mathbf{R}\mathbf{F}^{-1}\mathbf{R}'$ is positive definite because $\mathbf{R}$ has full row rank. By (v) and continuity of inversion, $\mathbf{R}\hat{\mathbf{G}}_n^{-1}\mathbf{R}' \xrightarrow{p} \mathbf{R}\mathbf{F}^{-1}\mathbf{R}'$. Now apply Theorem P.115. In the general case the same argument runs with the limiting variance $\mathbf{G}^{-1}\mathbf{F}\mathbf{G}^{-1}$ of (iii) and its consistent estimate. $\square$

Theorem P.114 (Likelihood ratio). Under Theorem P.112 with $\mathbf{F} = \mathbf{G}$, partition $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)$ with $\boldsymbol{\theta}_2$ of dimension q , and let $\hat{\boldsymbol{\theta}}_n^R$ maximise ℓ_n over $\Theta \cap \{\boldsymbol{\theta}_2 = \boldsymbol{\theta}_{2,0}\}$, where $\boldsymbol{\theta}_{2,0}$ is the true value. Suppose the hypotheses of Theorem P.112 also hold for the restricted problem, again with $\mathbf{F} = \mathbf{G}$ and with limiting information $\mathbf{F}_{11}$, the leading $(p - q)$ block of $\mathbf{F}$. Then

$$LR_n = 2\{\ell_n(\hat{\boldsymbol{\theta}}_n) - \ell_n(\hat{\boldsymbol{\theta}}_n^R)\} \xrightarrow{d} \chi_q^2.$$

Proof. Applying Theorem P.112(iv) to the full problem and, in the subspace $\{\boldsymbol{\theta}_2 = \boldsymbol{\theta}_{2,0}\}$, to the restricted one,

$$2\{\ell_n(\hat{\boldsymbol{\theta}}_n) - \ell_n(\boldsymbol{\theta}_0)\} = \mathbf{Z}'_n \mathbf{F}^{-1} \mathbf{Z}_n + o_p(1), \quad 2\{\ell_n(\hat{\boldsymbol{\theta}}_n^R) - \ell_n(\boldsymbol{\theta}_0)\} = \mathbf{Z}'_{n,1} \mathbf{F}_{11}^{-1} \mathbf{Z}_{n,1} + o_p(1),$$

where $\mathbf{Z}_n = n^{-1/2} \mathbf{U}_n \xrightarrow{d} Z \sim N(\mathbf{0}, \mathbf{F})$ and $\mathbf{Z}_{n,1}$ is its first $p - q$ coordinates, the score of the restricted model. Subtracting,

$$LR_n = \mathbf{Z}'_n \mathbf{F}^{-1} \mathbf{Z}_n - \mathbf{Z}'_{n,1} \mathbf{F}_{11}^{-1} \mathbf{Z}_{n,1} + o_p(1) \xrightarrow{d} Z' \mathbf{F}^{-1} Z - Z'_1 \mathbf{F}_{11}^{-1} Z_1$$

by the continuous mapping theorem and Slutsky.

Write $\mathbf{A} = [\mathbf{I}_{p-q} \quad \mathbf{0}]$, so $Z_1 = \mathbf{A}Z$ and $\mathbf{F}_{11} = \mathbf{AFA}'$, and put $W = \mathbf{F}^{-1/2}Z$, which is $N(\mathbf{0}, \mathbf{I}_p)$ by Theorem P.68 and the symmetric square root of Theorem P.41. Then $Z'\mathbf{F}^{-1}Z = |W|^2$ and $Z_1'\mathbf{F}_{11}^{-1}Z_1 = W'\mathbf{P}W$ with

$$\mathbf{P} = \mathbf{F}^{1/2}\mathbf{A}'(\mathbf{AFA}')^{-1}\mathbf{AF}^{1/2},$$

which is symmetric and, by the cyclic property of the trace,

$$\mathbf{P}^2 = \mathbf{P}, \quad \text{tr } \mathbf{P} = \text{tr}\{(\mathbf{AFA}')^{-1}\mathbf{AFA}'\} = p - q.$$

Hence $\mathbf{I} - \mathbf{P}$ is symmetric idempotent of rank q and the limit is $W'(\mathbf{I} - \mathbf{P})W \sim \chi_q^2$ by Theorem P.74. □

Remark. The three statistics use different parts of the same expansion: Wald uses $\hat{\boldsymbol{\theta}}_n$ and the curvature at it, the likelihood ratio uses the heights of the two fitted log-likelihoods, and both are χ_q^2 in the limit. They are not equal in finite samples, and Unit 7 exhibits a data set on which they disagree.

31 The Statistics of This Course

Theorem P.115 (Quadratic forms of asymptotically normal vectors). Let $V_n \xrightarrow{d} N(\mathbf{0}, \Sigma)$ in $\mathbb{R}^p$ with Σ positive definite, and let $\hat{\Sigma}_n \xrightarrow{p} \Sigma$. Then

$$V_n' \hat{\Sigma}_n^{-1} V_n \xrightarrow{d} \chi_p^2.$$

Proof. Matrix inversion is continuous on the open set of invertible matrices, and Σ lies in it, so $\hat{\Sigma}_n^{-1} \xrightarrow{p} \Sigma^{-1}$ by Theorem P.91. By Theorem P.92(i), $(V_n, \hat{\Sigma}_n^{-1}) \xrightarrow{d} (V, \Sigma^{-1})$ with $V \sim N(\mathbf{0}, \Sigma)$, and $(\mathbf{v}, \mathbf{A}) \mapsto \mathbf{v}' \mathbf{A} \mathbf{v}$ is continuous, so Theorem P.91 gives $V_n' \hat{\Sigma}_n^{-1} V_n \xrightarrow{d} V' \Sigma^{-1} V$.

It remains to identify the limit law. Let $\Sigma^{1/2}$ be the symmetric positive definite square root and $W = \Sigma^{-1/2} V$. By Proposition P.82 and Proposition P.81(iv), $W \sim N(\mathbf{0}, \mathbf{I}_p)$, so its coordinates are independent standard normals, and $V' \Sigma^{-1} V = |W|^2 = \sum_{j \leq p} W_j^2$, which is χ_p^2 by definition. $\square$

Remark (The Breusch–Pagan statistic). Unit 4 needs exactly this. There $V_n = n^{-1/2} \mathbf{Z}' \mathbf{g}$ is shown to satisfy $V_n \xrightarrow{d} N(\mathbf{0}, \kappa \mathbf{Q})$ by Corollary P.95: the summands $\mathbf{z}_i(\varepsilon_i^2/\sigma^2 - 1)$ are independent, mean

zero, and carry non-identical weights, which is why the Lindeberg–Feller form is required and the identically-distributed Corollary P.94 is not enough. The constant is $\kappa = 2$ under normality and $\kappa = \tau^2/\sigma^4$ in general. The empirical $n^{-1}\mathbf{Z}'\mathbf{M}_0\mathbf{Z}$ converges to $\mathbf{Q}$, so Theorem P.115 delivers the χ_p^2 limit, and Slutsky is what allows the unknown σ^2 and κ to be replaced by consistent estimates without changing the limit.

Remark (Where else). Units 7–8 use Theorem P.109 for the asymptotic distribution of functions of the maximum likelihood estimator, Theorem P.115 for the Wald statistic, and Theorem P.92 to substitute the observed information for its expectation. Unit 10 uses Theorem P.93 for the least squares estimator of an autoregression, where the summands $y_{t-1}\varepsilon_t$ are again not identically distributed.

32 Numerical Illustration

The Lindeberg condition is not a technicality: when it fails, the conclusion fails. The array below has $\sum_j \text{var}(\xi_{nj}) = 1$ for every n and mean zero, but one summand carries a fixed share of the variance, violating Step 1 of the proof of Theorem P.93.

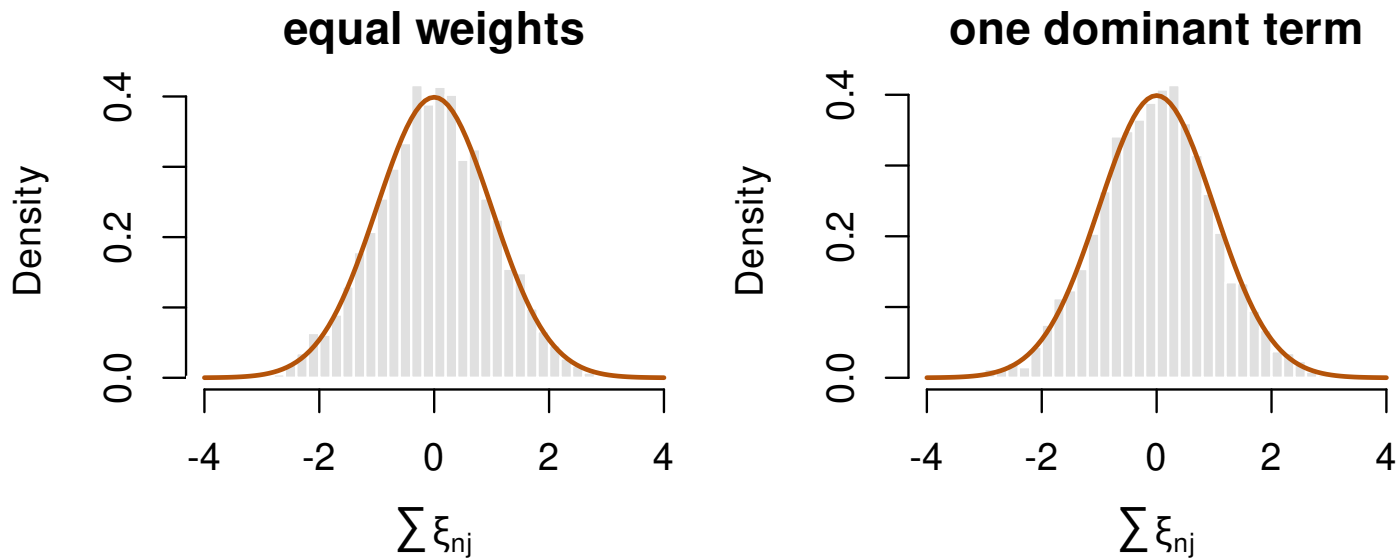
```

1 set.seed(1)
2 one_row <- function(n, share) {
3   # summand 1 carries `share` of the total variance; the rest split the remainder
4   v1 <- share; vr <- (1 - share) / (n - 1)
5   c(rnorm(1, sd = sqrt(v1)), rnorm(n - 1, sd = sqrt(vr)))
6 }
7 lindeberg_sum <- function(n, share, eps = 0.3, B = 4000) {
8   v1 <- share; vr <- (1 - share)/(n - 1)
9   sds <- c(sqrt(v1), rep(sqrt(vr), n - 1))
10  # E[xi^2 ; |xi|>eps] for a centred normal with sd s
11  contrib <- sapply(sds, function(s)
12    2 * integrate(function(x) x^2 * dnorm(x, sd = s), eps, Inf)$value)
13  sum(contrib)
14 }
15 c(share_0      = lindeberg_sum(400, 1/400),    # equal weights: condition holds
16   share_half   = lindeberg_sum(400, 0.5))      # one dominant term: condition fails

```

```
##          share_0      share_half
## 0.00000007488 0.49037575671
```

With equal weights the Lindeberg sum is numerically zero and the row sums are close to normal; with one dominant summand it is bounded away from zero and the row sums retain that summand's shape.



Corollary [P.95](#) is what the regression applications use. Its two design conditions are limits and so cannot be verified on a single sample; what a sample does show is the finite- n quantity behind the second of them, the largest share of $\sum_i |\mathbf{z}_i|^2$ carried by one observation. If that share is small the

Figure P.1: Row sums of a triangular array against the $N(0, 1)$ density. Both rows have total variance 1; only the left satisfies (P.19). In the right panel the sum is a sum of two normals and so is itself exactly normal. The illustration is that the limit is not reached **by accumulation**, the dominant term never becoming negligible.

design is far from the degenerate case, and if it is close to 1 no central limit theorem applies however large n is.

```

1 suppressMessages(library(ISLR2))
2 z  <- Boston$lstat; nn <- length(z)
3 Zc <- z - mean(z)                                # centred, as Z'MOZ requires
4 c(Q_n      = sum(Zc^2)/nn,                        # estimates Q; must be positive
5   max_share = max(Zc^2)/sum(Zc^2),                # largest single share
6   share_if_equal = 1/nn,                          # the value under equal weights
7   ratio      = max(Zc^2)/sum(Zc^2)*nn)            # how many times the equal share

```

##	Q_n	max_share	share_if_equal	ratio
##	50.893979	0.024889	0.001976	12.593774

The largest observation carries about 2.5% of the total, roughly twelve times an equal share: unequal,

but nowhere near domination by one point.

```

1  set.seed(3)
2  B <- 4000
3  Tn <- replicate(B, {
4    eps <- rnorm(nn)                                # H0: homoscedastic normal
5    g    <- eps^2 - 1
6    Vn   <- sum(Zc * g)/sqrt(nn)
7    Vn^2 / (2 * sum(Zc^2)/nn)                        # Theorem A.30 with p = 1
8  })
9  c(mean_stat = mean(Tn), chisq1_mean = 1,
10    var_stat  = var(Tn),  chisq1_var  = 2,
11    rejection_rate_5pct = mean(Tn > qchisq(0.95, 1)))

```

##	mean_stat	chisq1_mean	var_stat
##	1.03319	1.00000	2.09932
##	chisq1_var	rejection_rate_5pct	
##	2.00000	0.05375	

The simulated statistic has the mean, variance and upper tail of a χ_1^2 , which is Theorem P.115 with the covariance $\kappa\mathbf{Q}$ and $\kappa = 2$ supplied by Corollary P.95 under normality.

Theorem P.108 makes two claims that can be separated. The variance of $n^{-1/2} \sum_t u_{t-1} \varepsilon_t$ is $\tau^4 \sum_j a_j^2$ **exactly**, for every n , by Corollary P.107; what the theorem adds is that the law is asymptotically normal. Both are checked below for $a_j = \phi^j$ with $\phi = 0.8$, the score of a first-order autoregression, together with the truncation bound $\mathbf{E}[(S_n - S_n^{(m)})^2] = \tau^4 \sum_{j \geq m} a_j^2$ that drives the proof.

```

1 set.seed(5); phi <- 0.8; B <- 4000; ms <- c(1, 4, 16)
2 sim <- function(n) t(replicate(B, {
3   e <- rnorm(n + 80); k <- 81:(n + 80)
4   u <- as.numeric(stats::filter(e, phi, "recursive"))
5   S <- sum(u[k - 1]*e[k])/sqrt(n)
6   c(S, sapply(ms, function(m) {
7     um <- as.numeric(stats::filter(e, phi^(0:(m - 1)), sides = 1))
8     S - sum(um[k - 1]*e[k])/sqrt(n) })))
9 R <- sim(400)
10 t(sapply(seq_along(ms), function(i)
11   c(m = ms[i], mean_squared_difference = mean(R[, i + 1]^2),
12     theory_tail_sum = phi^(2*ms[i])/(1 - phi^2))))

```

```

##      m mean_squared_difference theory_tail_sum
## [1,] 1          1.802536          1.777778

```

## [2,]	4	0.436599	0.466034
## [3,]	16	0.002291	0.002201

The truncation error is what the approximation lemma needs to be small uniformly in n , and it is: at $m = 16$ the m -dependent array of the proof reproduces the score to within 0.0023 in mean square, against a variance of 2.78.

```

1 t(sapply(c(50, 200, 800), function(n) {
2   S <- sim(n)[, 1]
3   c(n = n, variance = var(S), theory = 1/(1 - phi^2),
4     skewness = mean((S - mean(S))^3)/sd(S)^3,
5     kurtosis = mean((S - mean(S))^4)/var(S)^2) })))

```

##	n	variance	theory	skewness	kurtosis
## [1,]	50	2.723	2.778	0.9377	4.755
## [2,]	200	2.835	2.778	0.5092	3.368
## [3,]	800	2.767	2.778	0.2295	3.136

The variance matches at every n , as it must. The skewness and kurtosis fall towards their normal values 0 and 3, roughly halving the gap each time n is quadrupled, which is the $n^{-1/2}$ rate. The approach is not fast: at $n = 50$ the kurtosis is still 4.76, so normality here is an asymptotic statement

and not a small-sample one.

33 Exercises

Exercise P.1. Give a sequence converging in distribution but not in probability, and one converging in probability but not almost surely.

Solution. For the first, let $X \sim N(0, 1)$ and $X_n = (-1)^n X$. Each X_n has the $N(0, 1)$ law, so $X_n \xrightarrow{d} X$; but $|X_n - X| = 2|X|$ for odd n , which does not tend to 0 in probability. This also shows convergence in distribution says nothing about the joint behaviour of X_n and X , only about laws.

For the second, take the sliding indicators on $[0, 1]$ with Lebesgue measure: for $n = 2^k + r$ with $0 \leq r < 2^k$, let $X_n = \mathbf{1}_{[r2^{-k}, (r+1)2^{-k}]}$. Then $\mathbf{P}\{X_n \neq 0\} = 2^{-k} \rightarrow 0$, so $X_n \xrightarrow{p} 0$; but every ω lies in infinitely many of the intervals, so $X_n(\omega)$ does not converge for any ω . ■

Exercise P.2. Show by example that $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$ do not imply $X_n + Y_n \xrightarrow{d} X + Y$, and identify which hypothesis of Theorem P.92 rules the example out.

Solution. Let $X \sim N(0, 1)$, $X_n = X$ and $Y_n = (-1)^n X$. Then $X_n \xrightarrow{d} N(0, 1)$ and $Y_n \xrightarrow{d} N(0, 1)$, yet $X_n + Y_n$ equals $2X$ for even n and 0 for odd n , so its law alternates between $N(0, 4)$ and δ_0 and does not converge.

Slutsky requires the second sequence to converge to a **constant**. That is exactly what fails: convergence in distribution constrains only marginal laws, and the sum depends on the joint law, which is unconstrained. Convergence to a constant pins the joint law down, which is the content of part (i) of the theorem. ■

Exercise P.3. Verify the Lindeberg condition for $\xi_{ni} = a_{ni}\varepsilon_i$ with ε_i i.i.d. mean zero, variance 1, where $\sum_i a_{ni}^2 = 1$ and $\max_i |a_{ni}| \rightarrow 0$.

Solution. $\mathbf{E}[\xi_{ni}] = 0$ and $\sum_i \text{var}(\xi_{ni}) = \sum_i a_{ni}^2 = 1$. Writing $\delta_n = \max_i |a_{ni}|$, the event $\{|\xi_{ni}| > \varepsilon\}$ is contained in $\{|\varepsilon_i| > \varepsilon/\delta_n\}$, so

$$\sum_i \mathbf{E}[\xi_{ni}^2; |\xi_{ni}| > \varepsilon] \leq \sum_i a_{ni}^2 \mathbf{E}[\varepsilon_1^2; |\varepsilon_1| > \varepsilon/\delta_n] = \mathbf{E}[\varepsilon_1^2; |\varepsilon_1| > \varepsilon/\delta_n],$$

using that the ε_i are identically distributed. Since $\varepsilon/\delta_n \rightarrow \infty$ and $\mathbf{E}[\varepsilon_1^2] = 1 < \infty$, dominated

convergence sends this to 0. The condition $\max_i |a_{ni}| \rightarrow 0$ is essential: with $a_{n1} = 1$ and the rest zero the sum is $\mathbf{E}[\varepsilon_1^2; |\varepsilon_1| > \varepsilon]$, which does not vanish. ■

Exercise P.4. Let $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \sigma^2)$ with $\theta > 0$. Find the limit law of $\sqrt{n}(\log \hat{\theta} - \log \theta)$ and of $\sqrt{n}(\hat{\theta}^2 - \theta^2)$.

Solution. By Theorem P.109 with $g(x) = \log x$, $g'(\theta) = 1/\theta$, so $\sqrt{n}(\log \hat{\theta} - \log \theta) \xrightarrow{d} N(0, \sigma^2/\theta^2)$: the asymptotic variance of a log-transformed estimator is the squared coefficient of variation.

With $g(x) = x^2$, $g'(\theta) = 2\theta$, giving $N(0, 4\theta^2\sigma^2)$. Note the method fails where $g'(\theta) = 0$: at $\theta = 0$ the second display would give a degenerate limit, and the correct normalisation is then n rather than $\sqrt{n}$, with a χ_1^2 limit. ■

Exercise P.5. Let $\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(\mathbf{0}, \Sigma)$ with Σ positive definite and let $\hat{\Sigma} \xrightarrow{p} \Sigma$. Show that

under $H_0: \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ with $\mathbf{R}$ of full row rank q ,

$$n(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'(\mathbf{R}\hat{\boldsymbol{\Sigma}}\mathbf{R}')^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}) \xrightarrow{d} \chi_q^2.$$

Solution. Under H_0 , $\sqrt{n}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}) = \mathbf{R}\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} N(\mathbf{0}, \mathbf{R}\boldsymbol{\Sigma}\mathbf{R}')$ by Theorem P.91 applied to the linear map $\mathbf{v} \mapsto \mathbf{R}\mathbf{v}$. The limiting covariance is positive definite: for $\mathbf{v} \neq \mathbf{0}$, $\mathbf{R}'\mathbf{v} \neq \mathbf{0}$ because $\mathbf{R}$ has full row rank, so $\mathbf{v}'\mathbf{R}\boldsymbol{\Sigma}\mathbf{R}'\mathbf{v} > 0$. Also $\mathbf{R}\hat{\boldsymbol{\Sigma}}\mathbf{R}' \xrightarrow{p} \mathbf{R}\boldsymbol{\Sigma}\mathbf{R}'$ by Theorem P.91. Theorem P.115 with $p = q$ then gives the χ_q^2 limit. This is the Wald statistic of Units 7–8. ■

Exercise P.6. In Theorem P.93 normality is nowhere assumed of the ξ_{nj} , yet the proof introduces Gaussian variables ζ_{nj} . Explain their role, and say where in the Breusch–Pagan derivation of Unit 4 normality of the **errors** does real work.

Solution. The ζ_{nj} are a comparison array, not an assumption: they are constructed to have the same variances c_{nj} as the ξ_{nj} , so that their sum is exactly $N(0, c_n)$ and the problem reduces to showing the two products of characteristic functions are close. Normality enters only as a device for producing a reference sum with a known law.

In Unit 4 normality of the errors is used in exactly one place: to evaluate $\text{var}(\varepsilon_i^2/\sigma^2) = 2$, which fixes the divisor 2 in $LM = \text{SSR}_z/2$. Without it that variance is τ^2/σ^4 and the correctly scaled statistic is Koenker's nR_z^2 . The central limit theorem behind both is the same, and needs no normality at all. ■